Welcome
to

GENERAL PSYCHOLOGY COURSE

Ground Rules

- *1 hour lecture followed by student questions and answers*
- *15 minute break (get a snack!)*
- *Another 1 hour lecture followed by student questions and answers*
- *Then the end of class for today*

UNIT ONE

ESSENCE OF PSYCHOLOGY

Overview

- ***Definition of Basic Concepts***
- ***Goals of Psychology***
- ***Historical Background of Psychology***
- ***Major Perspective in Psychology***
- ***Branches of Psychology***
- ***Research Methods in Psychology***

Meaning and Definition of Psychology

- The word “Psychology "comes from the two Greek words. These are:

✓ ***psyche***, which translates as **“soul”** or “sprit”, ”mind” and

✓ ***logos***, which means the **study,** knowledge or discourse.

“ the study of the **mind/soul/sprit**”

represented by the Greek letter ψ (psi) which is read as("sy")

psychology *is a scientific study of behavior and its underlying mental process of human beings and animals.*

Key words in the definition

Science

- ✓ Follow scientific procedures and use empirical data to study behavior and mental processes.
- ✓ Psychology does not rely on common sense or speculation

Behavior

- ✓ All of our out ward or overt actions and reactions ,such as talking, facial expressions, movement ,etc.
- ✓ There is also covert behavior which is hidden, non-observable and generally considered as a mental process

Cont…

Mental processes

- ✓ Refer to all the internal, covert activities of our minds, such as thinking, feeling, remembering, etc.

- ✓ Psychologists strive to understand the ***mysteries of human nature***—why people **think**, **feel**, and **act** as they do

Psychologists also study **animal behavior**;

It purposes:

- ✓It is **ethically forbidden** to conduct experiment on human beings.
- ✓ Conclusions obtained from experiments on animal behavior are usually **applicable** to human behavior
- ✓ To formulate **theories**, **laws** & **principles** that govern human behavior
- ✓To determine **laws of behavior** that apply to all organisms

Review

- "Scientific study of behaviour and its causes."
  - Overt (directly observable) and covert behaviours
- Psychologists study:
  - How you act (behaviour/overt)
  - How you think (mental/covert)
  - How you feel (covert & overt)
  - How your brain and body respond (physiological/covert)

Goals of Psychology

- ✓ **Description**: what is happening?
- ✓ **Explanation**: why it is happening?
- ✓ **Prediction**: When will it happen again?
- ✓ **Controlling**: How can it be changed?

Description

- ✓ it involves observing the behavior and noticing everything about it.
- ✓ Every behaviour has its own way of occurring
- ✓ In describing behaviour, a psychologist focuses on how behaviour occurs.
- ✓ It is a search for answers for questions like
- ✓ What is happening? Where does it happen? To whom does it happen?' And under what circumstances does it seem to happen?.

Explanation

- ✓ In explanation of behaviour, a psychologist becomes concerned about **why behaviour occurs as it does**
- ✓ Every behaviour has its own causes. No behaviour occurs without a cause.
- ✓ It is about trying to **find reasons** for the observed behavior.
- ✓ This helps in the process of forming theories of behavior (A theory is a general explanation of a set of observations or facts).

Prediction

- ✓ Prediction is about determining **what will happen in the future**
- ✓ involves forecasting the likelihood of a behaviour under certain circumstances.
- ✓ Prediction of behaviours is possible through the use of **theories or principles**

Control (Modification)

- ✓ **How can it be changed?** Control or modify or change the behavior from undesirable one to a desirable one).
- ✓ involves changing a behaviour which is anti social or unacceptable.
- ✓ For healthy functioning of society and the individual, these kind of negative (maladaptive) behaviours should be avoided
- ✓ In psychology, there are psychological techniques to help an individual avoid a maladaptive behaviour.

Historical Roots of Psychology

- Psychology has its roots in **<u>philosophy</u>** and **<u>physiology</u>**
- **<u>Philosophers</u>** had asked questions about human emotions, thoughts and behavior. They had tried to deduce answers to their questions by applying logic and common sense reasoning philosophers did not always make deduction successfully.
- **who contributed to the Development of Psychology**
  - ✓Hippocrates (460-377 B.C)- emotion(body humor)
  - ✓Plato (428-348 B.C)_intelligence (inborn/inherited)
  - ✓Aristotle (382-322 B.C)_thinking (heart)
  - ✓Rene Descartes (1556-1650 A.D)_mind/body
  - ✓John Locke (English Philosopher)_tabula rasa

- ✓ **Empiricists** (a group of philosophers who believed a pursuit of truth through observation and experience)
- ✓ **Nativism** (group of philosophers who believed a knowledge is inborn or inherited)

- **Physiologists** were especially influential in providing a new understanding of the brain and the nervous system, and the way in which these structures affect behavior.

It was the union between the questions asked by the philosophers and the careful scientific experimentation of the physiologists that led to the field of study we call psychology.

Major Perspective in Psychology

Early Schools of Psychology

Structuralism

- developed by Edward Bradford Titchener.
- Wilhelm Wundt, who established the first scientific laboratory of psychology in Leipzig in 1879, and believed that human mind could be scientifically studied.
- Task of Psychology
  - is to **identify the basic elements of consciousness (image, feelings & sensation)**
  - to find out the units or elements,which make up the mind
- Methods: ***Introspection*** *(*looking in ward into our consciousness*)*

Functionalism

- founded by William James (1848-1910) which proposed that, the *function of the mind*, not the structure.

Task of Psychology

✓is to investigate the function, or purpose, of consciousness rather than its structure

✓psychological processes are **adaptive**. They allow humans to survive and to adapt successfully to their surroundings.

Method

✓questionnaires, mental tests and objective descriptions of behavior

Gestalt psychology

- Founders of this schools of thought are
  - ✓ *Max Wertheimer (1880-1943)*

Task of Psychology

✓mind should be thought of as resulting from the whole pattern

✓Psychology as a study of the **whole mind**

✓Argued that the mind is not made up of combinations of elements.

✓The mind should be thought of as a result of the whole pattern of sensory activity and the relationships and organizations within their pattern

Methods

✓ ***are Naïve Introspection and experimentation***

Behaviorism

- Founder-**John B. Watson** (1878-1958)

Task of psychology

✓Behaviorists view psychology as a study of observable and measurable **behaviors**.

✓Three important characteristics; conditioned response, learned rather than unlearned behaviors, and focus on animal behavior

Methods

- are Observation and Experimentation

Psychoanalysis

- founded by a Viennese physician Sigmund Freud (1856-1939).

Task of Psychology

✓ psychology studies about the components of the unconscious part of the human mind.

✓The unconscious which is the subject matter of psychoanalys is contains hidden wishes, passions, guilty secrets, unspeakable yearnings, and conflict between desire and duty.

Methods

- free association, dream interpretation, analysis of slip of tongue, jokes, and Transference

The mind is like an iceberg in that only a small part of its substance is visible

Modern Perspectives in Psychology

Psychodynamic perspective

- ✓ It has its origins in ***Freud's theory*** of psychoanalysis, but many other psychodynamic theories exist.
- ✓ This perspective emphasizes the ***unconscious dynamics*** within the individual such as inner forces, conflicts or instinctual energy.
- ✓ The psychodynamic approach emphasizes:
  - The influence of unconscious mental behavior on every day behavior
  - The role of ***childhood experiences*** in shaping adult personality
  - The role of ***intrapersonal conflict*** in determining human behavior
- Psychodynamic perspective tries to dig below the surface of a person's behavior to get into unconscious motives
- Psychodynamists think of themselves as ***archaeologists of the mind.***

Behavioral Perspective

- It emphasizes the role *learning experiences* play in shaping the behavior of an organism.
- It is concerned with how the environment affects the person's actions.
- Behaviorists focus on environmental conditions(e.g. rewards, and punishments) that maintain or discourage specific behaviors.
- Also called the "*black box*" approach in psychology because it treats the mind as less useful in understanding human behavior and focus on what goes into and out of the box, but not on the processes that take place inside
- This means, behaviorists are only interested in the effects of the environment (input) on behavior (output) but not in the process inside the box.

Humanistic Perspective

- ✓ Human behavior is not determined either by unconscious dynamics or the environment.
- ✓ Rather it emphasizes the *uniqueness of human beings* and focuses on human values and subjective experiences.
- ✓ This perspective places greater importance on the individual's ***free will.***
- ✓ The goal of humanistic psychology was helping people to *express themselves creatively and achieve their full potential or self- actualization* (developing the human potential to its fullest).

Cognitive Perspective

- It emphasizes what goes on in people's **heads**; how people reason, remember, understand language, solve problems, explain experiences and form beliefs.
- This perspective is concerned about the **mental processes.**
- The most important contribution of this perspective has been to show *how people's thoughts and explanations affect their actions, feelings, and choices*.
- Techniques used to explore behavior from a cognitive perspective include *electrical recording of brain activity, electrical stimulation and radioactive tracing of metabolic activity in the nervous system*.

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Biological Perspective

- It focuses on studying how bodily events or *functioning of the body* affects behavior, feelings, and thoughts.
- It holds that the *brain and the various brain chemicals* affect psychological processes such as learning, performance, perception of reality, the experience of emotions, etc.
- This perspective underscores that *biology and behavior interact in a complex way*; biology affecting behavior and behavior in turn affecting biology.
- It also emphasizes the idea that we are ***physical beings*** who evolved over along time and that genetic heritage can predispose us to behaving in a certain way.

Socio-cultural Perspective

- It focuses on the *social and cultural factors* that affects human behavior.
- Cultural psychologists also examine how cultural *rules and values* (both explicit and unspoken) affect people's development, behavior, and feelings.
- This perspective holds that humans are both the products and the producers of culture, and our behavior always occurs in some cultural contexts.

Branches of Psychology

Psychology is a broad field, there are many specialization under its umbrella

Developmental Psychology

- ✓ Studies how people develop *overtime* thorough the process of maturation and learning.
- ✓ studies age related changes through the life span
- ✓ Aspects of Development(Physical, Cognitive, Social, etc)
- ✓ Stages of Development (Infancy, Babyhood, childhood, adolescence, adulthood, old age)
- ✓ It attempts to examine the major developmental milestones that occur at different stages of development.

Cont…

Educational Psychology

- Concerned with the application of psychological principles and theories in improving the educational process including curriculum, teaching, and administration of academic programs.

Counselling Psychology

- Helps individuals with less severe problems than those treated by clinical psychologists.

✓ assists people on issues of personal adjustment, vocational and career planning, family life and may work in schools, hospitals, clinics or offices

Cont…

Personality Psychology

- ✓It focuses on the relatively enduring traits and characteristics of individuals.
- ✓Study topics such as self-concept, aggression, moral development, etc.
- ✓studies individual differences in personality and their effects on behaviour

Industrial(Organizational) Psychology

- ✓ Studies human behaviour in the workplace and how behaviour affects production
- ✓ Applies *psychological principles* in industries and organizations to increase the productivity of that organization.

Cont…

Social Psychology

- ✓ It studies the role of social forces in governing individual behaviour.
- ✓ Examines the ways in which the pattern of a person's feeling, thinking and acting is affected by others
- ✓ Deals with people's social interactions, relationships, social perception, and attitudes.

Cont...

Cross-cultural Psychology

- ✓ Examines the role of culture in understanding behavior, thought, and emotion.
- ✓ It compares the nature of psychological processes in different cultures, with a special interest in whether or not psychological phenomena are universal or culture-specific.

Forensic psychology

- ✓ Applies psychological principles to improve the legal system (police, testimony,etc..).

Cont…

Health Psychology

- ✓ Applies psychological principles to the prevention and treatment of physical illness and diseases.

Clinical Psychology

- ✓ Is a field that applies psychological principles to the prevention, diagnosis, and treatment of psychological disorders.

Research Methods in Psychology

Definition of Terms

- Research: Is a scientific method of gathering and testifying data by applying different methods and making conclusion and prediction of the phenomenon.
- **Theory-** is a statement which is generalized from scientific study and which explains the relationship among variables.
- It is an integrated set of principles helpful to organize, explain and predict events.

- **Hypothesis** : is any statement or assumption that serves as a possible but tentative explanation of certain observation.

- It is an educated guess that can be tested.

- It is a statement of cause and effect relationship.

- It is useful to guide a study.

Variables : are constructs that vary or change. There are two events or constructs. The variation of one construct may be followed by the variation of another construct.

- **Population** is a group of subjects (universe) under study.

For example, children under 5 year of age; primary school children in Sidama Zone.

- **Sample** is a small portion of a population that is expected to be representative of the population (universe). Sample is better required than population for different reasons

Cont…

Scientific method- a process of testing ideas through systematic observations, experimentations, and statistical analysis.

Theory-is an integrated set of principles about observed facts that is intended to describe and explain some aspects of experience.

Hypotheses-is a tentative proposition about the relationship between two or more variables or phenomena.

E.g. Males have high self-confidence in making decisions than females.

Major types of research methods

Descriptive research methods

- ✓ In this type of research, the researcher simply records what she/he has systematically observed.
- ✓ Include naturalistic observation, case studies, and surveys.

Correlational research methods

- ✓ Is are search method that measures the relationship between two or more variables

Experimental Research

- ✓ It is are search method that allows researchers to study the cause and effect relationship between variables

1. Naturalistic Observation

- A researcher engages in **careful observation** of behavior **without intervening** directly with the subjects.
- A research method in which various aspects of behavior carefully observed in the setting where such behavior **naturally occurs**.
- it allows researchers to study behavior under conditions that are **less artificial** than in experiments.

2. Case Study

- is an **in-depth** investigation of an individual subject.
- is an intensive study of a person or group. Most case studies combine long-term observations with **diaries**, **tests**, and **interviews**.

3. Survey

- use **questionnaires or interviews** to gather information about specific aspects of participants' background and behavior.
- One of the most practical ways to gather data on the **attitudes, beliefs, and experiences of large numbers of people** is through **surveys.**

4. Longitudinal Studies

- It studies the ***same group of people at*** regular intervals over a period of years to determine whether their behavior and/or feelings have changed and if so, how.

5. Cross-Sectional Studies

- People studied from **different age** groups at same time point.

6. Correlations

- studying the **relationship** between two variables such as between weight and height, chewing chat and score, and age and academic achievement.
- The **correlation coefficient** is a numerical index of the degree of relationship between two variables. A correlation coefficient indicates

  (1) the direction (positive or negative) of the RXnship

  (2) how strongly the two variables are related.

(1)

- ***A positive correlation*** indicates that two variables co-vary in the *same* direction.
- **A *negative* correlation** indicates that two variables co-vary in the *opposite* direction.

(2)

- the ***size of the coefficient*** indicates the ***strength*** of an association between two variables. The coefficient can vary between 0 and 1.00 (if positive) or between 0 and 1.00 (if negative).
- A coefficient near 0 indicates no relationship between the variables.

7. Experimental Method

- allows researchers to detect cause-and-effect relationships.
- the investigator manipulates a variable under carefully controlled conditions and observes whether any changes occur in a second variable as a result.
- There are *two types* of variables: independent and dependent.

- ***The independent variable :*** is a condition or event that an experimenter varies in order to see its impact on another variable.
- ***The dependent variable:*** is the variable that is thought to be affected by manipulation of the independent variable.

Example

1) the number of hours you study affects your performance on an exam.
2) the effect of watching violence TV program on children behavior.

- In an experiment the investigator typically assembles two groups of subjects who are treated differently with regard to the independent variable. These two groups are referred to as the experimental group and the control group.

a) The **experimental group** consists of the subjects who receive some special treatment in regard to the independent variable.
b) The **control group** consists of similar subjects who do not receive the special treatment given to the experimental group.

Steps of scientific research

Step one- Defining the Problem

- ✓ Noticing something attention catching in the surrounding for which one would like to have an explanation.

Step two-Formulating the Hypothesis

- ✓ after having an observation on surroundings (perceiving the problem),you might form an educated guess about the explanation for your observations, putting it into the form of a statement that can be tested in someway.

Cont..

Step three-Testing the Hypothesis

- ✓ At this step, the researcher employs appropriate research methods and collects ample data (information) to accept or reject the proposed statement.

Step four- Drawing Conclusions

- ✓ This is the step in which the researcher attempts to make generalizations or draw implications from tested relationship

Step five-Reporting Results

- ✓ At this point, the researcher would want to write up exactly whats he/he did, why she/he did, and what she/he found.

Reading Assignment

- What were the chief tenets of structuralism and functionalism?
- What did Freud have to say about the unconscious and sexuality, and why were his ideas so controversial?
- What was the main idea underlying behaviorism?
- How do clinical psychology and psychiatry differ?
- Why study psychology? Why is psychology important for medicine?

See you next week…

& Happy Reading!

CHAPTER TWO

SENSATION AND PERCEPTION

Meanings of Sensation and Perception

Brainstorming questions

Have you heard of sayings like„

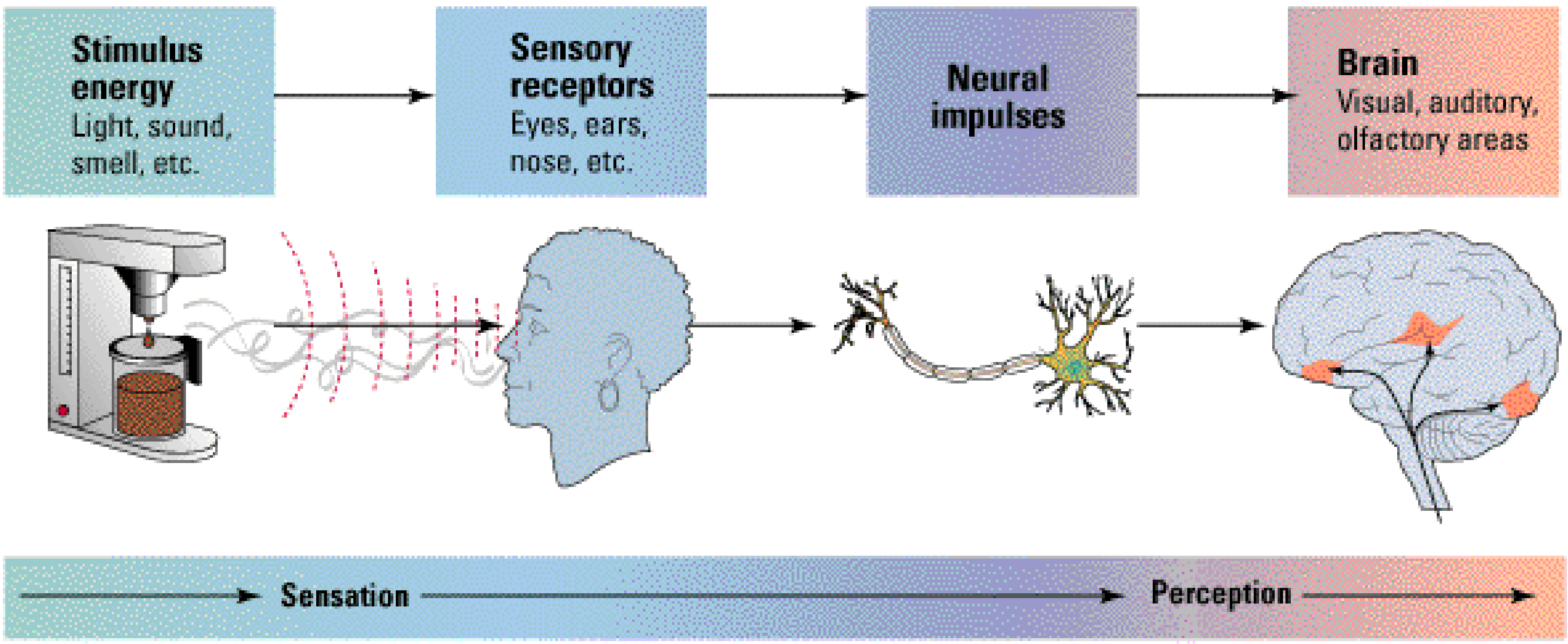
- *you watch but you don't see;*
- *you hear but you don't listen;*
- *you touch but you don't grasp...*

Which one do you think refers to sensation and which one refers to perception?

Basic Terms

- Stimulus
- Response
- Sensation
- Transduction
- Perception
- Psychophysics

Sensation and perception



Transduction

- Communication between the brain & the rest of the body (& between different regions of the brain) occurs via neuron. We recently learned how communication between neurons occurs *electrochemically* (within neurons: electrical; between neurons: chemical). So the brain's "language" is electrochemical!

- All senses involve something called *receptor cells*. Their job is to transduce (transform or even "translate") physical stimulation/physical energy from the environment into electrochemical messages that can be understood by the brain.

Sensation

- Process where by *stimulation* of receptor cells in the sensory systems sends nerve impulses to the brain.

e.g. Color, brightness, the pitch of tone or a bitter taste

- The starting of point of sensations is a **stimulus**. A form of energy (such as light waves or sound waves) that can affect **sensory organs** (such as the eye or the ear).

Therefore,

it is the process that **detects the stimulus** from one‘s body or from the environment.

Perception

- Process that organizes sensations into meaningful patterns.
- Process by which the brain selects, organizes, and interprets these **sensations**
- a meaning making process
- Or process where by the brain interprets sensations, giving them order and meaning.
- Thus, hearing sounds and seeing colors is largely a sensory process, but forming a melody and detecting patterns and shapes is largely a perceptual process.

The sensory laws

Sensory threshold

- ✓ Is the minimum point of intensity a sound can be detected.
- ✓ There are two laws of sensory threshold:
  - ➢ law of absolute threshold
  - ➢ law of difference threshold

The absolute threshold

- The **minimum a mount of stimulation** a person can detect
- As the minimum level of stimulation that can be detected **50 percent** of the time when a stimulus is presented over and over again.
- Thus, if you were presented with a low intensity sound 30 times and detected it 15 times, that level of intensity would be your absolute threshold for that stimulus.

Absolute thresholds

Vision

A single candle flame from 30 miles /48 km on a clear night

Hearing

The tick of a watch from 20 feet/6 meter in total quiet

Smell

One drop of perfume in a 6-room apartment

Touch

The wing of a bee on the cheek, dropped from 1 cm

Taste

One teaspoon of sugar in 2 gallons /7.7 liters of water

Difference Threshold

- The minimum amount of **change** that can be detected
- Or minimum change in stimulation that can be detected 50 percent of the time by a given person.
- Also, called **Just Noticeable Difference**
- Smallest difference that can be detected when 2 stimuli are compared.

e.g., you would have to increase the intensity of the sound from your tape recorder a certain amount before you could detect a change in its volume.

Sensory Adaptation

- if a stimulus remains **constant** in intensity, you will gradually **stop noticing** it
- tendency of our sensory receptors to have **decreasing responsiveness** to unchanging stimulus
- But, potentially important change in your environment while ignoring unchanging aspects of it.

Sensory Adaptation

Diminished sensitivity as a consequence of constant stimulation.

Put a band aid on your arm and after awhile you don't sense it.

Sensory Adaptation v Habituation

Sensory Adaptation

- Neural or sensory receptors change/reduce their sensitivity to a continuous, unchanging stimuli
- This occurs in the brain
- e.g. adapting to hot or cold water after a brief time in it
- e.g. the eyes adjusting to a darker room—rods and cones will fire differently to adjust (cones take about 10 min, the rods 30 min to fully adapt)
- The smell of your own house

Habituation

- Pattern of decreased response to a stimulus after frequently repeated exposure
- Occurs in the body
- Reduced response to something that used to elicit a stronger response
- e.g. response to drugs
- e.g. No longer respond to favorite food as when we first "loved" it

Attributes of Sensation

Sensory Deprivation- is the absence of normal level of sensory stimulation.

- Human brain requires a minimum amount of sensory stimulation in order to function normally. If it is too low it is bad for the brain to function properly.

Sensory Overload- is experiencing too much amount of stimulus from the environment. This is also bad for the brain to function properly.

- Generally too little stimulation (sensory deprivation) and too much stimulation (sensory overload) can lead to fatigue and mental confusion.

Perception

Major **characteristics** of the perceptual process:

- *selectivity of perception*
- *form perception,*
- *depth perception,*
- *perceptual constancy, and*
- *perceptual illusion.*

Selectivity of Perception: Attention

- Sense organ is bombarded by many stimuli..... perceive a few of them..... ignoring the other unnecessary stimuli.....ATTENTION
- **Attention**
  - It is perceptual process that selects certain inputs for inclusion in your conscious experience, or awareness, at any given time, ignoring others.

Factors Influencing Perceptual Selectivity

External Attention Factors

- Intensity
- Size
- Contrast
- Repetition
- Motion
- Novelty and
- familiarity

Internal Set Factors

- Learning
- Expectations
- Motivation
- Personality

What Affects Attention

- **Intensity-** the more intense the stimulus the more it'll be attended. E.g. the brighter light is more attended than the dull one.

- **Size-** the larger the size of the stimulus the more we give attention and the smaller the size we give less attention.

- **Contrast-** what contrasts with the surrounding environment attracts attention easily. E.g. if one stranger and teacher are enter in the class, the students give more attention to the stranger and less attention for the teacher.

- **Movement-** something, which moves, is more likely to attract attention than something stationary.
- **Motivation-** largely our current level of satisfaction or deprivation determines what we choose to hear or watch.
- **Personality and interest-** individuals select the stimulus and give attention if they are interested.

E.g. in the football game, a person may give attention to the game his wife may give attention to the music in the stadium.

Internal(Psychological) Factors that affect Attention

1. Set or Expectancy

- ✓ refers to mental readiness to receive certain kinds of sensory input

EX: A husband expecting an important phone call is more likely to hear phone calls than a wife who is concerned about her baby crying.

2. Motives or Needs

- ✓ People are more likely to be attracted to environmental experiences (events) in which they are interested

Form Perception

- The meaningful shapes or patterns or ideas that are made perhaps out of meaningless and discrete or pieces and bites of sensations

Perception has organization and structure

- ✓ Everything we perceive has its own **structure** and form. To make sense out of what we perceive, we must know where one thing begins and another ends
- ✓ This process of dividing up the world occurs effortlessly (naturally) and makes our perception more **meaningful**

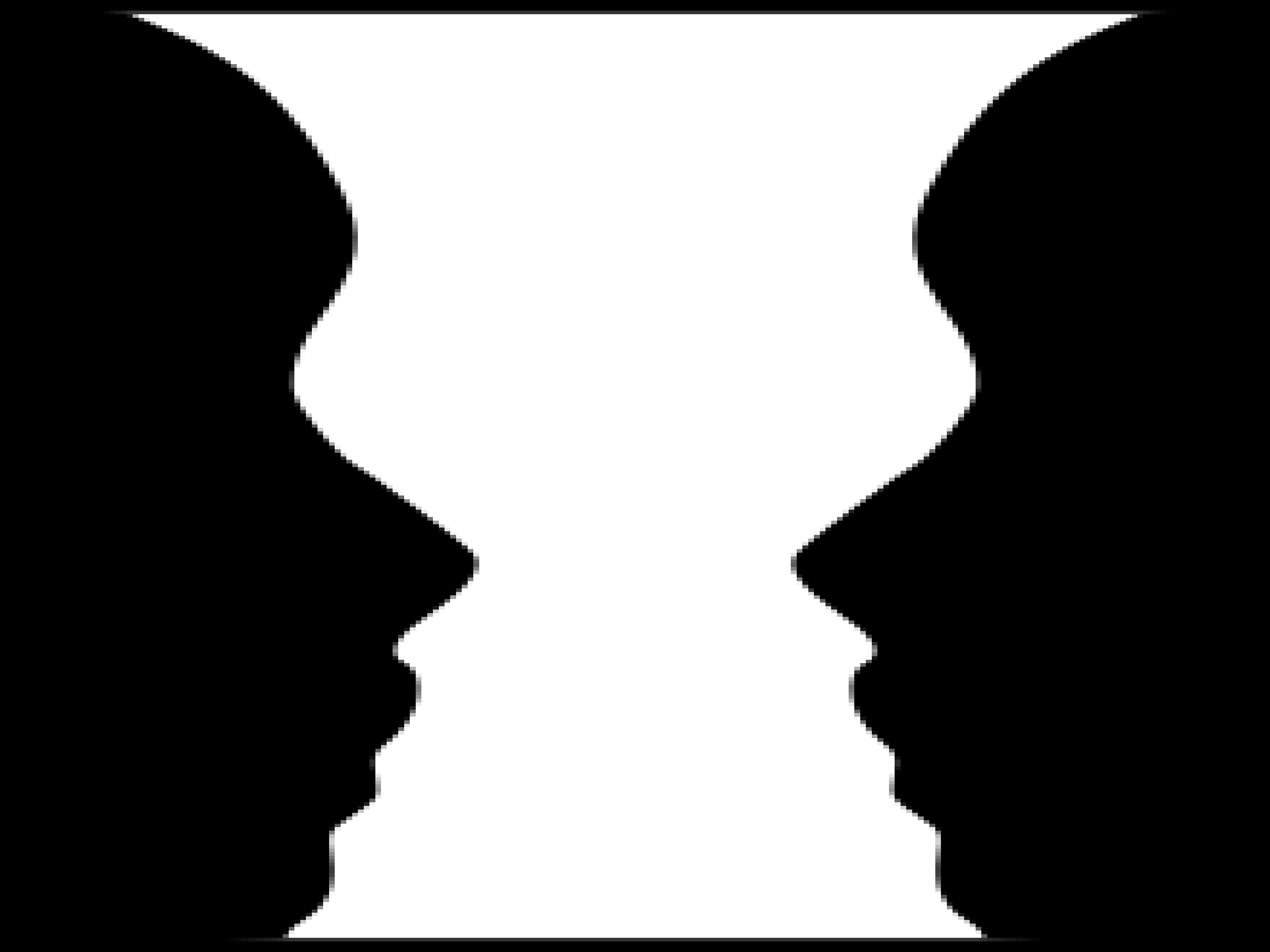
Principles (laws) of perceptual organization

- ✓ The **brain uses structures** in order to give pattern, shape and form to our visual perception.
- ✓ It is based on these principles that **perceptual organization** becomes possible.
- ✓ The gestalt laws of organization are principles that describe how we organize and construct pieces of information into meaningful wholes.

Gestalt psychologist said -***the whole is more than the sum of its parts.***

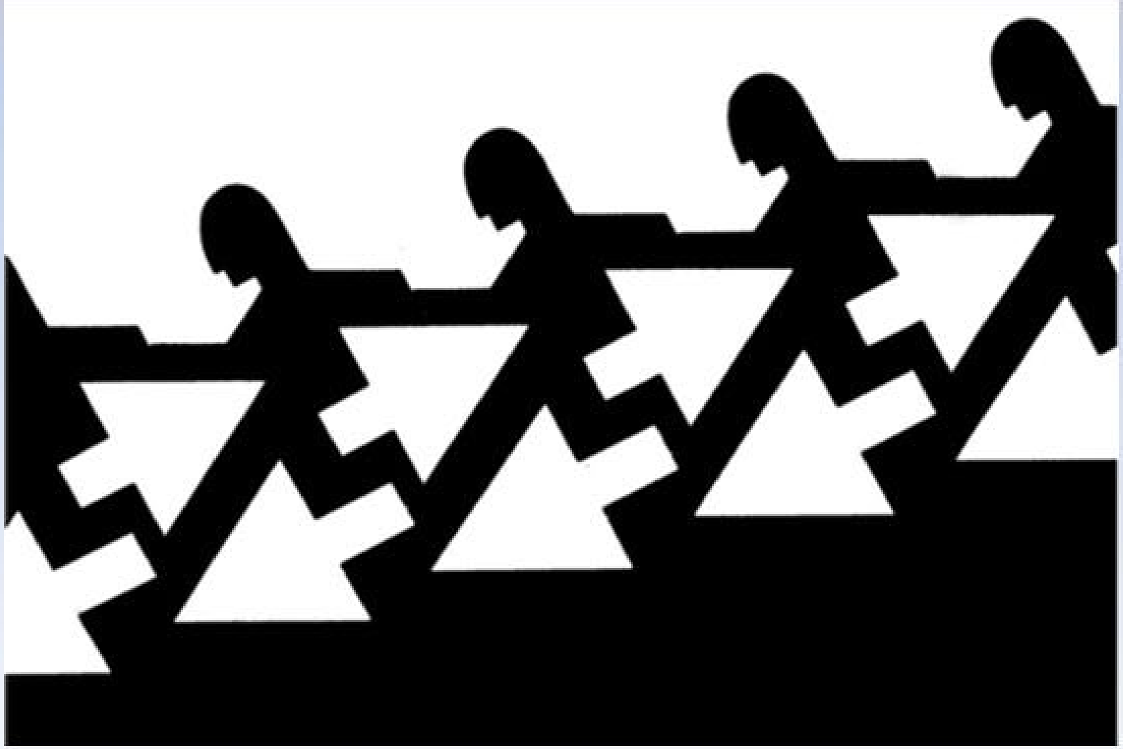
Figure-Ground Perception

- the perception of objects and forms of everyday experience as standing out from a background.
- This is a principle by which we organize the perceptual field in to stimuli that stand out (the figure) and those that are left over (the ground).



A. Form Perception

- ____________________: organization of the visual field into objects (the figures) that stand out from their surroundings (the ground).



A reversible figure-ground illustration

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The principle of Closure

- ✓ This is a principle that states the brain tends to fill in gaps in order to perceive complete forms.
- ✓ People need to decipher less than perfect images to make perceptions. To help us do so, the brain tends to *finish what is unfinished, complete what is incomplete.*

The principle of Proximity

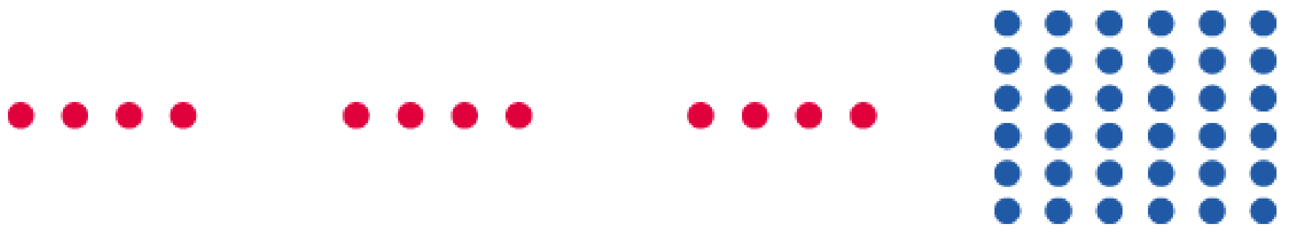
- ✓ This principle states that things that are *near each other tend to be grouped together*. The closer objects or events are to one another, the more likely they are to be perceived as belonging together.

Form Perception:

Gestalt principles

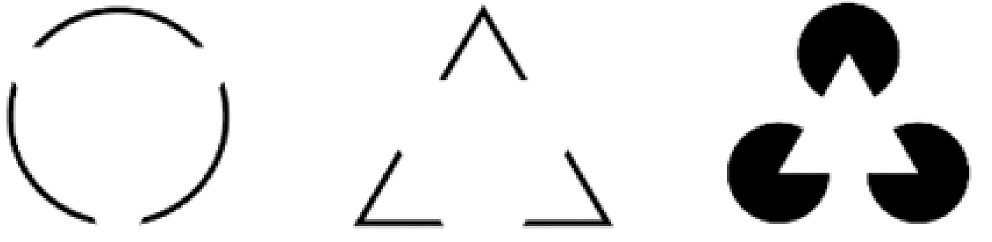
Proximity

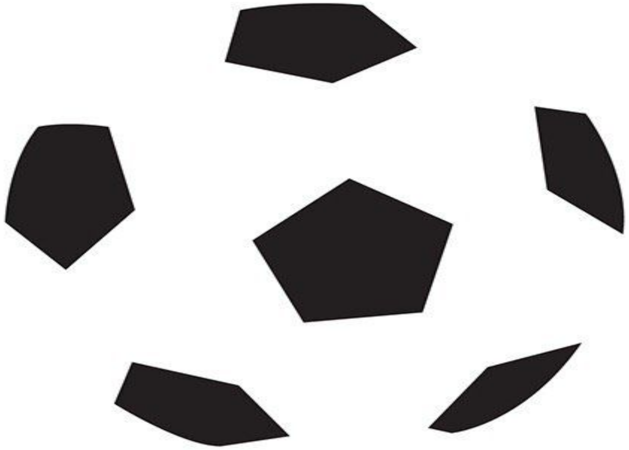
Things close to one another are grouped together



Closure

The brain tends to fill in gaps to perceive complete forms





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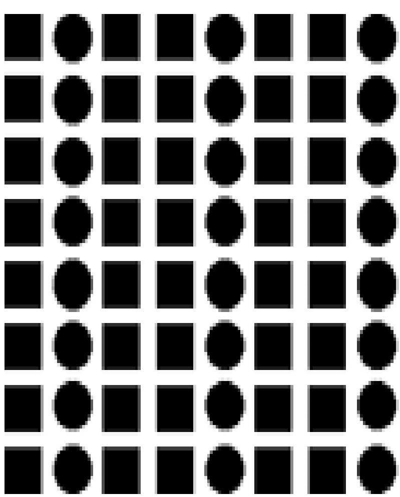
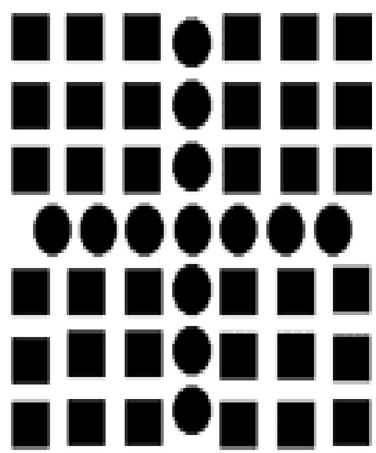
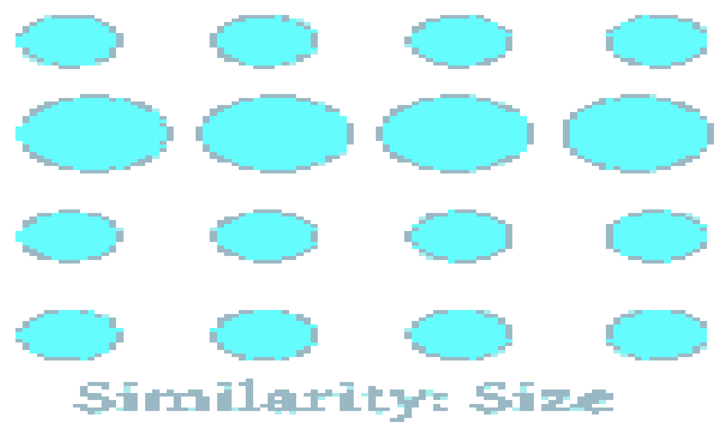
The principle of Similarity

- ✓ The principle of similarity states that things that are ***alike*** in some way (for example, in colour, shape or size) tend to be perceived as belonging together.
- ✓ Things that are alike are perceived together







Perception is Constant under Changing Sensory Information

- ✓ perception does ***not change*** when sensory information about stimuli changes.
- ✓ Our perceptual hypothesis remains the ***same*** when information we receive about stimuli through the visual sense organ change in colour, size or shape.

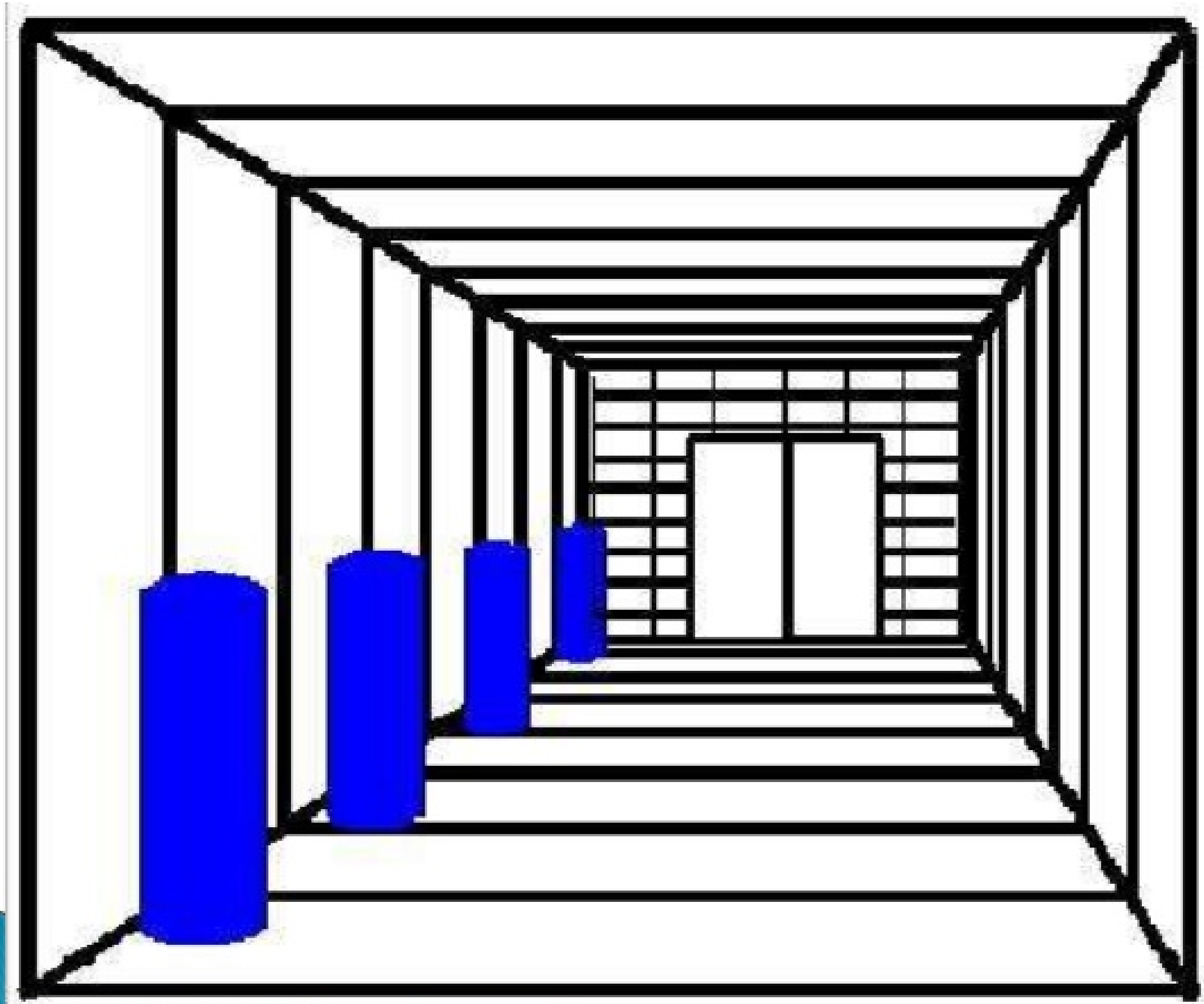
Categories of Perceptual Constancy

1. Size Constancy

- ✓ refers to the perception that the size of objects remains ***constant*** even though visual information change with variations in distance.

Size Constancy

Size constancy refers to our ability to see objects as maintaining the same size even when our distance from them makes things appear larger or smaller.



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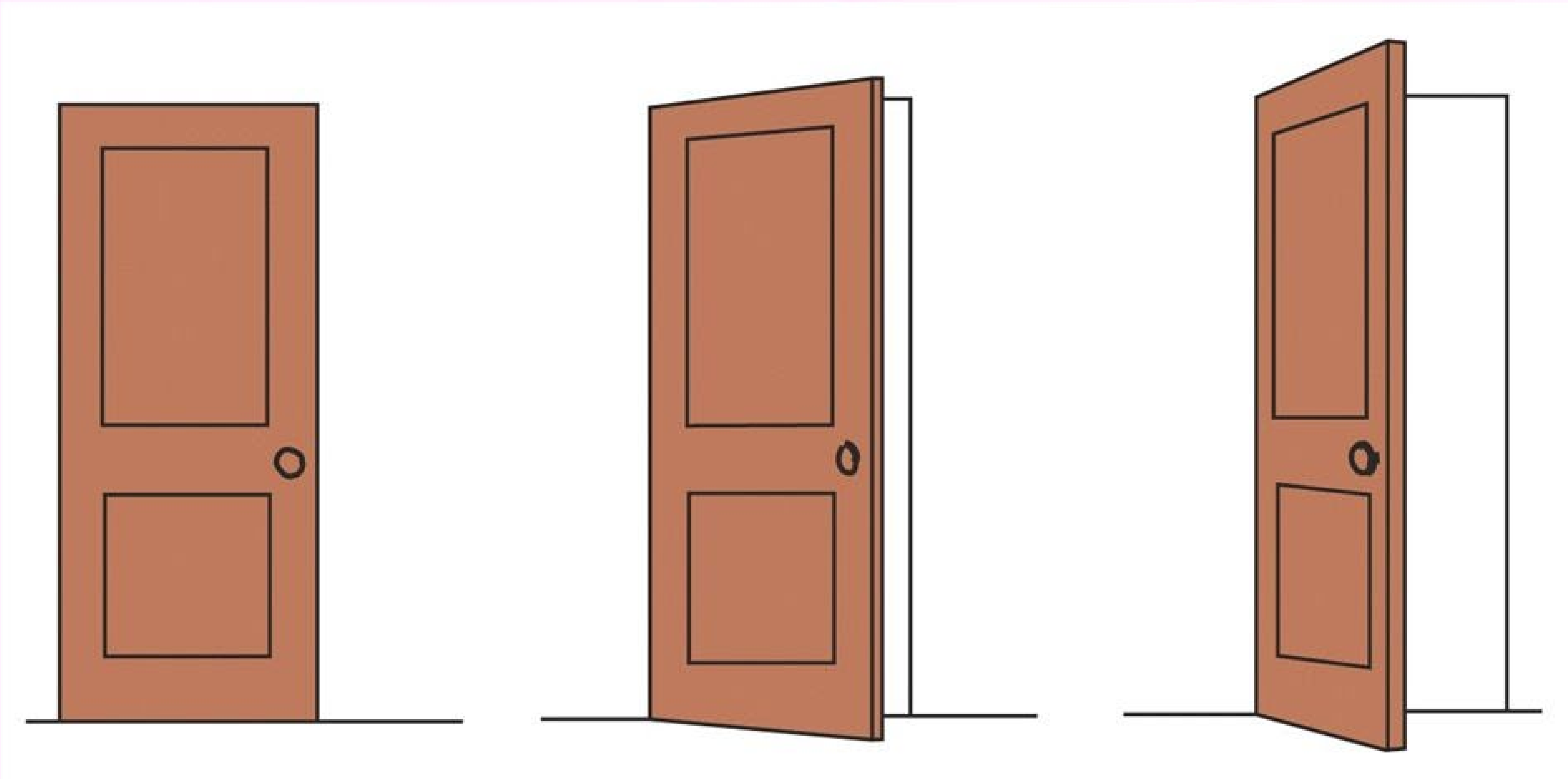
2. Shape Constancy

- ✓ states that we continue to perceive objects as having a ***constant shape*** even though the shape of the retinal image changes when our ***point of view changes***.
- ✓ Viewing angle or position superficially changes the shape of an object

Shape Constancy

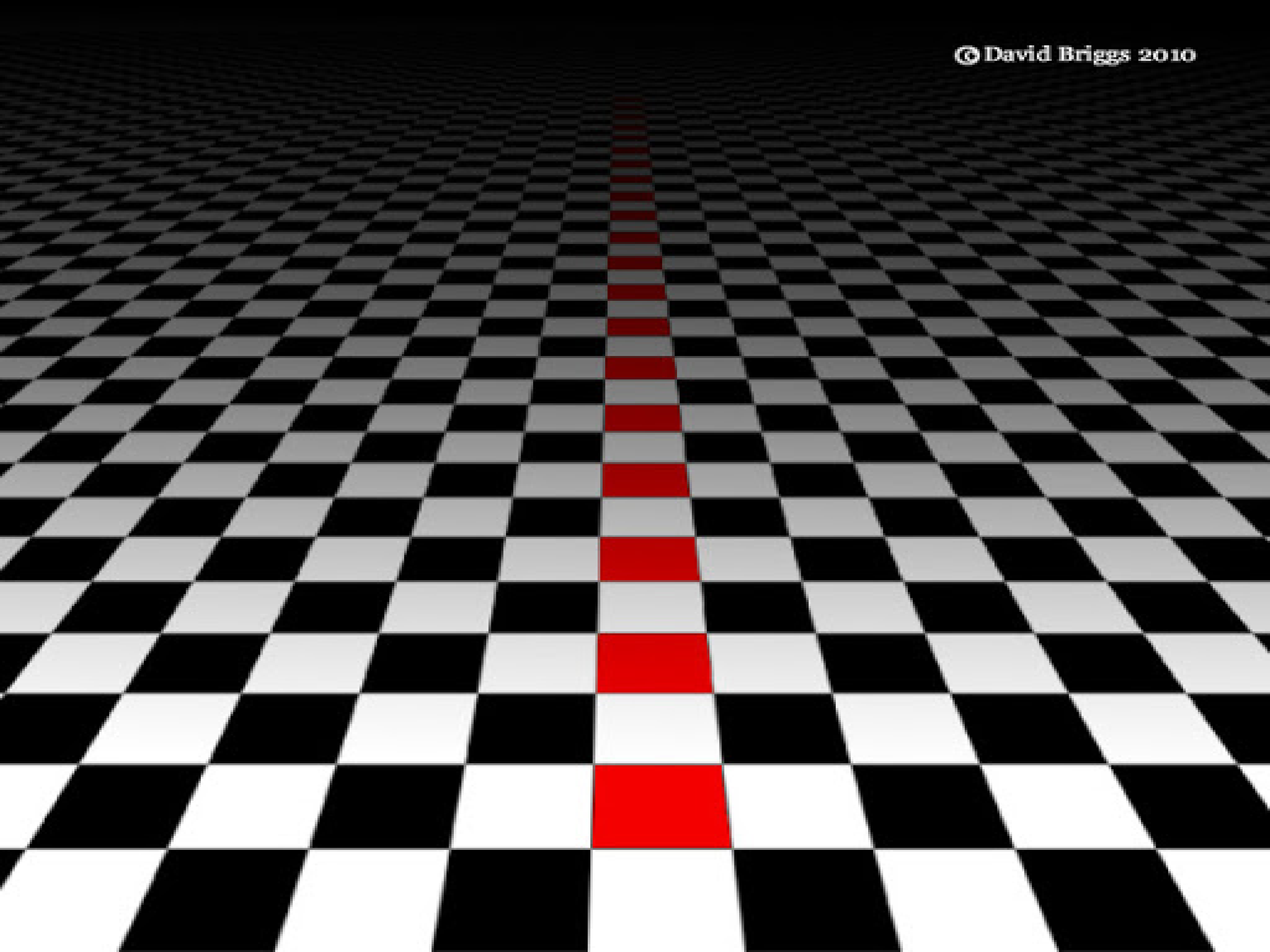
- The understanding that an object's shape remains the same even though the *angle* of view makes the shape appear changed



Cont...

3. Colour (Brightness) Constancy

- ✓ Sometimes objects may take different colour or brightness because of variations in light reflected on them.
- ✓ This principle states that the colour or brightness of an object remains the same even though the amount of light reflected on the objects change.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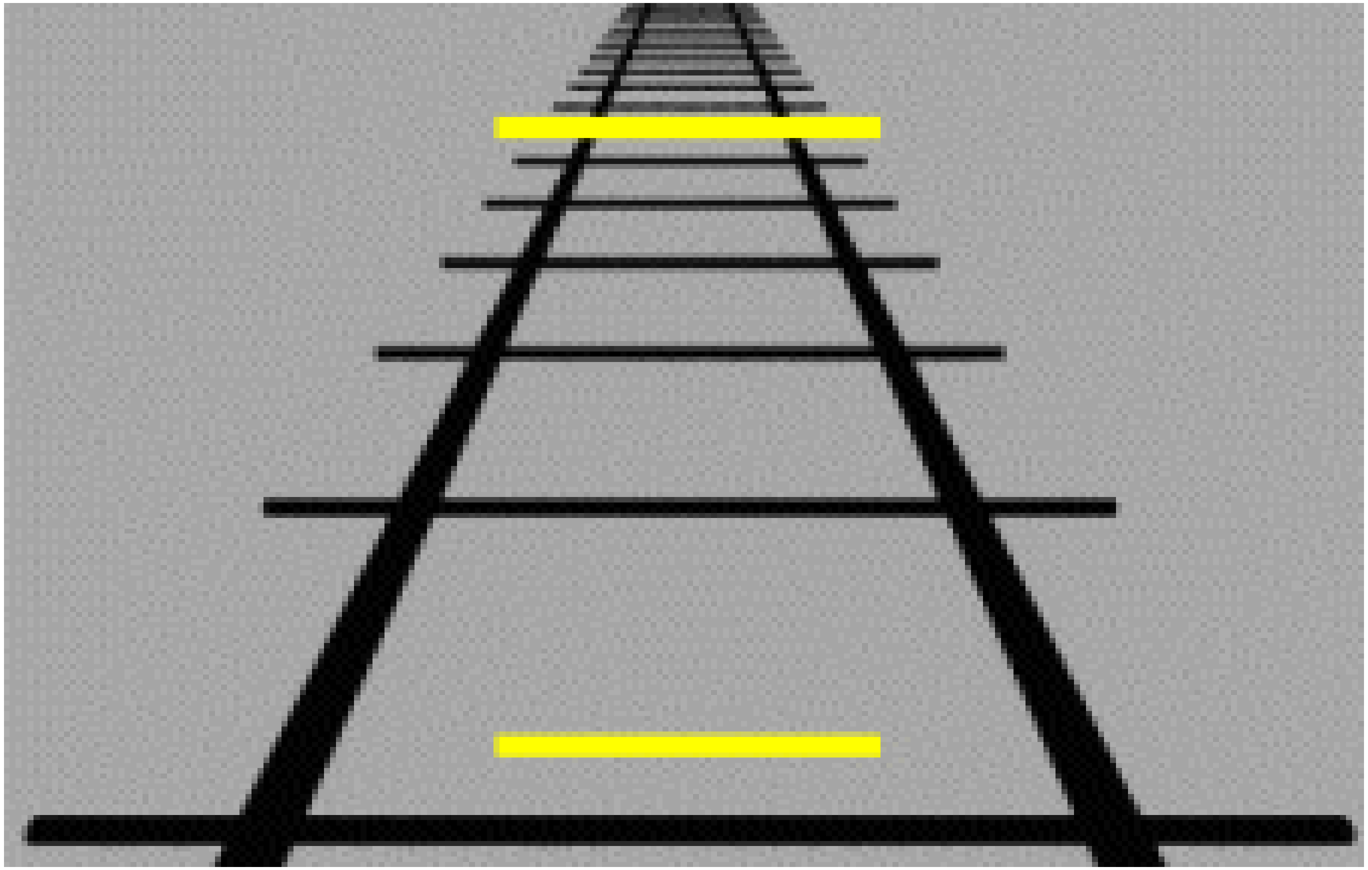
is affected by single light source.

Fig. 2: Example of Grey World

4. Location Constancy

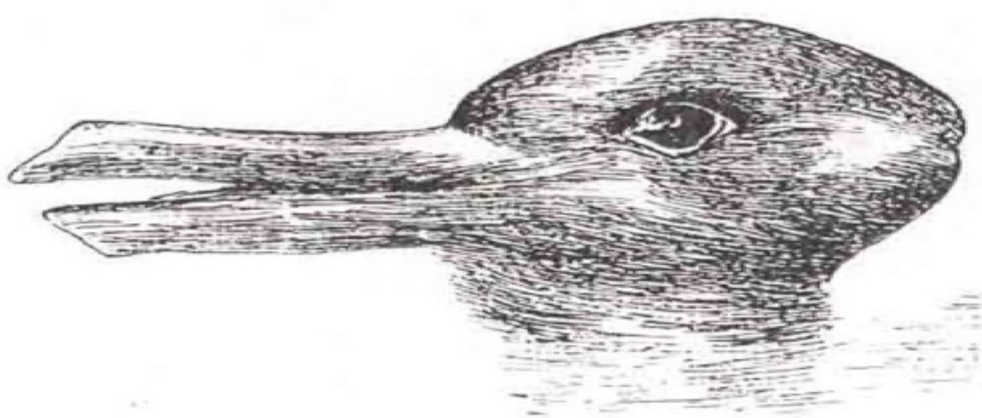
- ✓ The location or position of stationary objects is always the same even when our ***eyes tell us it is moving.***
- ✓ We perceive ***stationary objects as remaining*** in the same place even though the retinal image moves about as we move our eyes, heads, and bodies.



Location constancy refers to the relationship between the viewer and the object. A stationary object is perceived as remaining stationary despite the retina sensing the object changing as the viewer moves (due to parallax).

Perception illusion

- **Illusions** are special **perceptual** experiences in which information arising from “real” external stimuli leads to an incorrect **perception**, or false impression, of the object or event from which the stimulation comes.

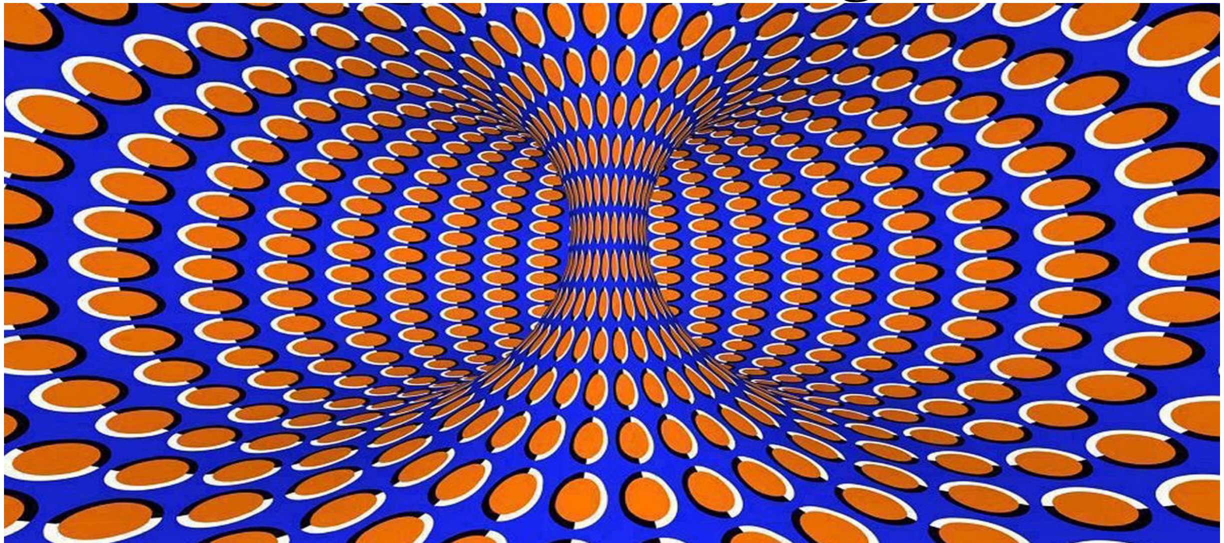


Auditory illusions

- Are *false perceptions* of a real ***sound*** or outside stimulus.
- The listener hears either sounds which are not present in the stimulus, or sounds that should not be possible given the circumstance on how they were created

Optical Illusions

- can use color, light and patterns to create images that can be deceptive or misleading to our brains.
- The information gathered by the eye is processed by the brain, creating a perception that in reality, does not match the true image.



Reading Assignment

- Explain the implications of sensation and perception in medical practice.
- Find an example of a perceptual illusion, it can be related to optical, auditory, or any sensory illusion. After sharing the illusion, you need to explain the process of how our brains perceive the stimuli.

See you next week…

& Happy Reading!

Chapter Three

Learning

Brainstorming Questions

- *What is the meaning of learning to you?*
- *What are the elements of learning?*
- *How do we learn?*

Discuss in Pair

- *Almost all human behavior is learned. Imagine if you suddenly lost all you had ever learned. You would be unable to read, write, or speak. You couldn't feed yourself, find your way home, and drive a car, play a game, or "party."*

- *What could you do?*

Discussion Points

- *What are the characteristics of learning?*
- *What does it take for learning to take place effectively?*
- *What do you think are the factors the affect your learning?*

“Learning is the eye of the mind”

“universal and distinctive characteristics of human beings is their capacity to learn”

Learning

is a relatively permanent change in behavior which is a function of prior experience or practice;

- ✓a relatively permanent change
- ✓change in behavior
- ✓depend on experience or practice

✓The learning is not directly observable but manifests in the activities of the individual.

Characteristics of learning

- continuous ***modification*** of behavior throughout life
- ***pervasive***, it reaches into all aspects of human life.
- involves the whole person, socially, emotionally & intellectually.
- often a change in the organization of experiences.
- responsive to ***incentives***
- an active process and purposeful
- depends on maturation, motivation and practice.
- multifaceted

Principles of Learning

- Individuals learn best when they are physically, mentally, and emotionally ***ready*** to learn.
- Students learn best and retain information longer when they have meaningful ***practice and exercise***
- Learning is **strengthened** when accompanied by a **pleasant** or satisfying feeling, and that learning is **weakened** when associated with an **unpleasant** feeling.
- Things ***learned first*** create a strong impression in the mind that is difficult to erase.
- Things most ***recently learned*** are best remembered.

Cont…

- The principle of ***intensity*** implies that a student will learn more from the **real thing** than from a substitute.
- Individuals must have **some abilities and skills** that may help them to learn.
- Things **freely learned** are best learned - the greater the freedom enjoyed by individuals, the higher the intellectual and moral advancement.

Factors Influencing Learning

- Motivation
- Maturation
- Health condition of the learner
- Psychological wellbeing of the learner
- Good working conditions
- Background experiences
- Length of the working period
- Massed and distributed learning

Theories of Learning

- Theories of learning attempt to explain the mechanism of behavior involved in the learning process. Generally, learning theories can be categorized as:

1. **Behavioral Learning Theories**
2. **Cognitive Learning Theories**
3. **Social Learning Theory**

Behaviorist Learning Theories

<u>Assumption</u>

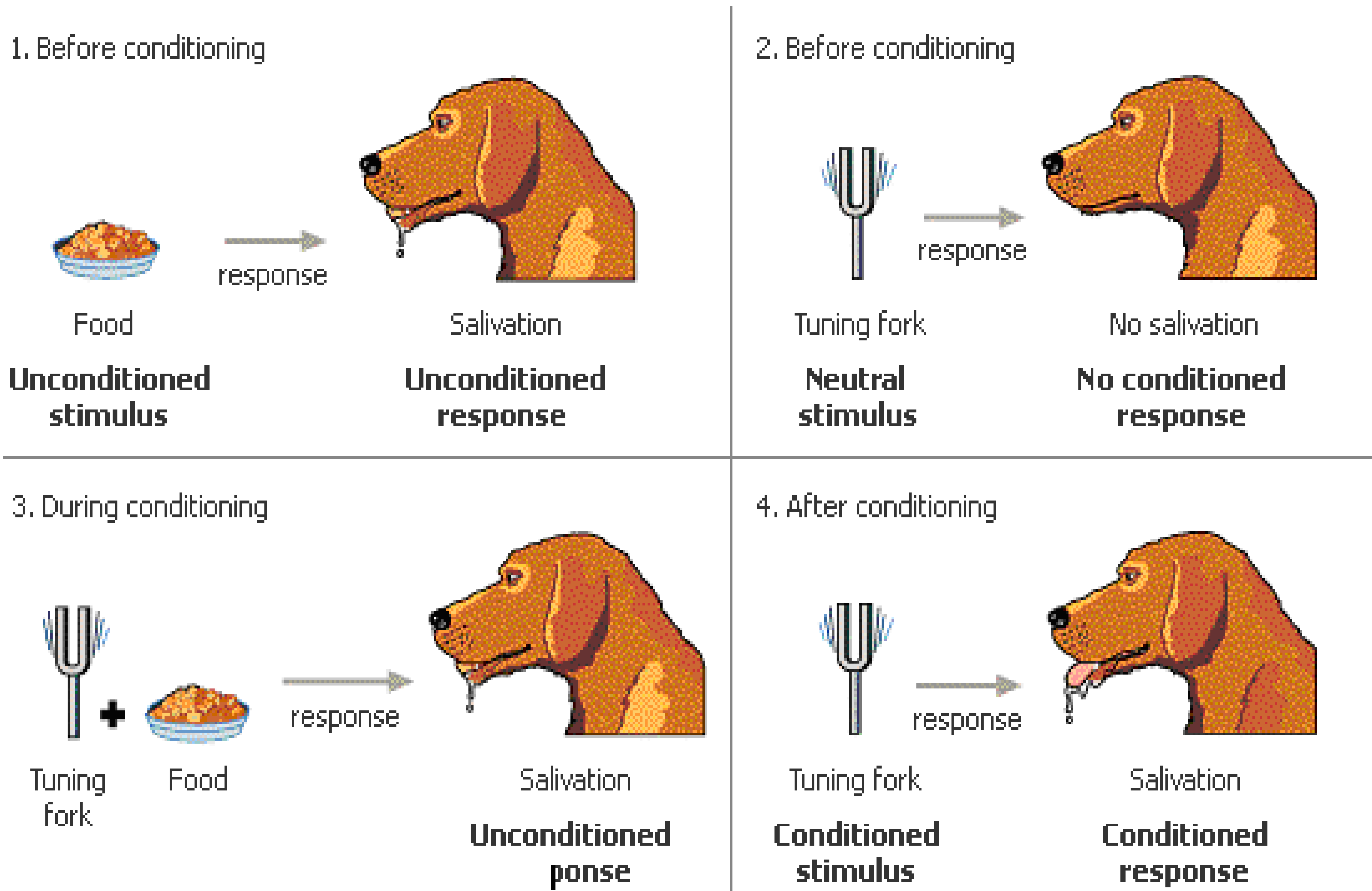
- ✓ learning as the product of the association between stimulus conditions (**S**) and the responses (**R**).
- ✓ The learner has to do some thing (respond to a stimulus) for learning to occur.
- ✓ ***Repetition*** of the stimulus-response (S-R) connection promotes learning
- ✓ Behavioral theories emphasize ***observable behaviors***, seek laws to govern all organisms, and provide explanations which focus on consequences.
- ✓ The ***<u>consequences</u>*** that follow the response of the learner to a stimulus can hinder or encourage learning
- ✓ Learning is verified through ***observation***

Classical Conditioning Theory of Learning

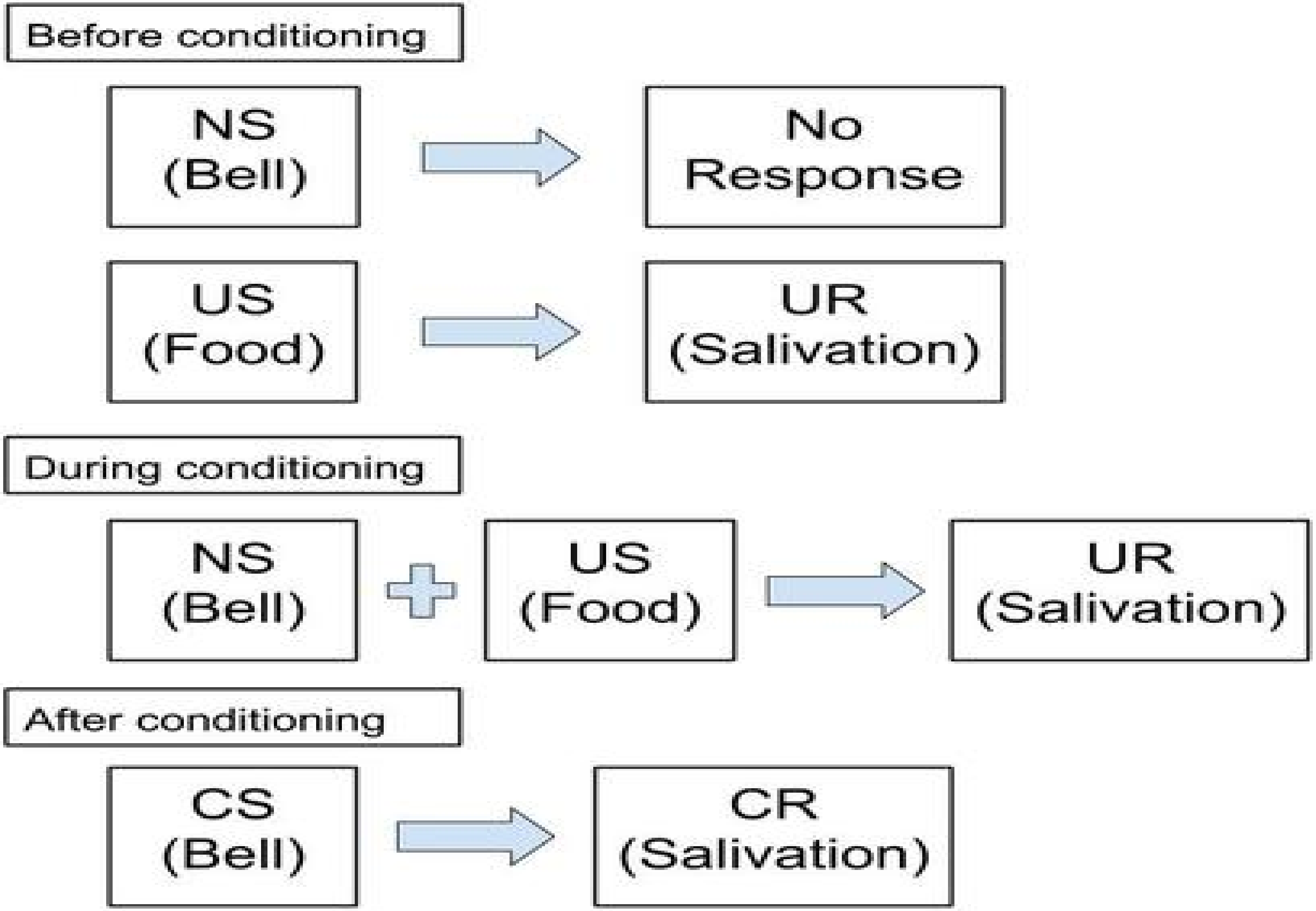
- ✓ founded by a Russian physiologist, Ivan Pavlov((1849-1936).).
- ✓ He studied the process of salivary secretion in dog.
- ✓ learning of involuntary emotional or physiological responses
- ✓ stimulus conditions and the associations formed in the learning process.

Definition

- ✓ Classical conditioning refers a learning situation in which a neutral/ conditioned stimulus gradually gain the ability to elicit a response because of its former paring with a natural/ unconditioned stimulus
- ✓ In the experiment, Pavlov identified three steps in the process of salivary conditioning which can be summarized as follows.


1. Before conditioning
Food
response
Salivation
Unconditioned stimulus
Unconditioned response
2. Before conditioning
Tuning fork
response
No salivation
Neutral stimulus
No conditioned response
3. During conditioning
Tuning fork
Food
response
Salivation
Unconditioned ponse
4. After conditioning
Tuning fork
response
Salivation
Conditioned stimulus
Conditioned response

Cont…



Before conditioning
NS (Bell)
No Response
US (Food)
UR (Salivation)
During conditioning
NS (Bell)
US (Food)
UR (Salivation)
After conditioning
CS (Bell)
CR (Salivation)

Elements of classical conditioning

- As we can see from the above procedure, classical conditioning as a paradigm involves four elements

Unconditioned stimulus (US)

Unconditioned stimulus is any event that elicits a natural response prior to conditioning. In the above case meat is the unconditioned stimulus because it caused salivation, automatically before conditioning took place.

Unconditioned response (UR)

Unconditioned response is the response to the natural unconditioned stimulus. The dog's salivation after receiving meat is the unconditioned response. Thus, unconditioned response is an automatic, involuntary and unlearned response to a particular stimulus is called unconditioned response

Conditioned stimulus (CS)

Conditioned stimulus is the previously neutral stimulus that acquired the power to elicit a response when it is associated with unconditioned stimulus. In the earlier experiment, the bell is a conditioned stimulus since, it latter became capable to make the dog to salivate

Conditioned response (CR)

Conditioned response is the learned response that is evoked by the conditioned stimulus. The dog's salivation in response to sound of the bell (in the absence of meat) is a conditioned response thus conditioning is said to be complete when the conditioned stimulus causes a conditioned response to occur with out the presentation of the unconditioned stimulus.

Example

Before Conditioning
Offensive odor in the hospital unwanted feeling by a visitor

During Conditioning
Hospital + Offensive odors unwanted feeling
Several pairings of hospitals
+
Offensive odors in hospitals unwanted feeling

After Conditioning
Hospital unwanted feeling

Basic Principles of Classical Conditioning

1. ***Acquisition:*** is a process in which a conditioned stimulus gradually acquires the capacity to elicit a conditioned response as the result of repeated pairing with an unconditioned stimulus. It also refers to initial learning of the CS gain power over the UCS to produce conditioning.

2. Principle of Association/Contiguity: states that a stimulus and response become connected if they occur close together in time and space.

3. ***Extinction:*** refers to the decline of CR in absence of UCS. It is actually inhibition of the CR rather than elimination of it.

4. ***Spontaneous Recovery:*** refers to the reappearance of CR after a rest pause.

5. ***Stimulus Generalization:*** refers to the tendency to react or respond to stimuli that are different from but some what similar to a conditioned stimulus.

6. ***Stimulus Discrimination:*** refers to the learning ability to distinguish between CS and other similar but irrelevant stimuli that do not signal conditioned stimulus. It is responding to CS but not to other similar stimuli.

Example

1. Visitors who feel discomfort in one hospital subsequently go to other hospitals to see relatives or friends without smelling offensive odors, and then their discomfort and anxiety about hospitals may be lessened after several such pairings.
2. It is so difficult to give up unhealthy habits, and addictive behaviors such as smoking, alcoholism or drug abuse. A cured addict of khat or alcohol may suddenly relapse to chewing or consuming if he/she is continuously exposed to khat or alcohol.
3. When listening to friends and relatives tell about a hospital experience, it becomes apparent that highly positive or negative personal encounters may color patients evaluations of their hospital stay as well as their subsequent feelings about being hospitalized in the future.

Reflection

- Suppose a one-year old child is playing with a toy near an electrical out-let. He sticks part of the toy into the outlet. He gets shocked, becomes frightened, and begins to cry. For several days after that experience, he shows fear when his mother gives him the toy and he refuses to play with it. What are the UCS? UCR? CS? CR? Show in diagram there association into three stages of processes?

a) UCS______________________________

b) UCR______________________________

c) CS______________________________

d) CR______________________________

Could you please explain of something you learned through classical conditioning?

Operant Conditioning Theory of Learning

✓ developed by an American psychologist, B. F. Skinner.

Assumptions

- learning in which a **voluntary response** is strengthened or weakened, depending on its favorable or unfavorable consequences
- **Environmental consequences** is at the heart of Operant Conditioning (also called *Instrumental Conditioning*)
- **Operation** or actions which an organism has to carry out
- Behavior can be explained by external causes of an action and the action‘s consequences.

- Operant conditioning theory of learning is concerned with voluntary and higher learning rather than reflexive or involuntary behavior.
- The term operant conditioning refers to the fact that the learner must operate, or perform a certain behavior, before receiving a reward or punishment.
- **Thus, by definition, operant conditioning is a type of learning in which behavior is strengthened if followed by reinforcement, or diminished if followed by punishment.**

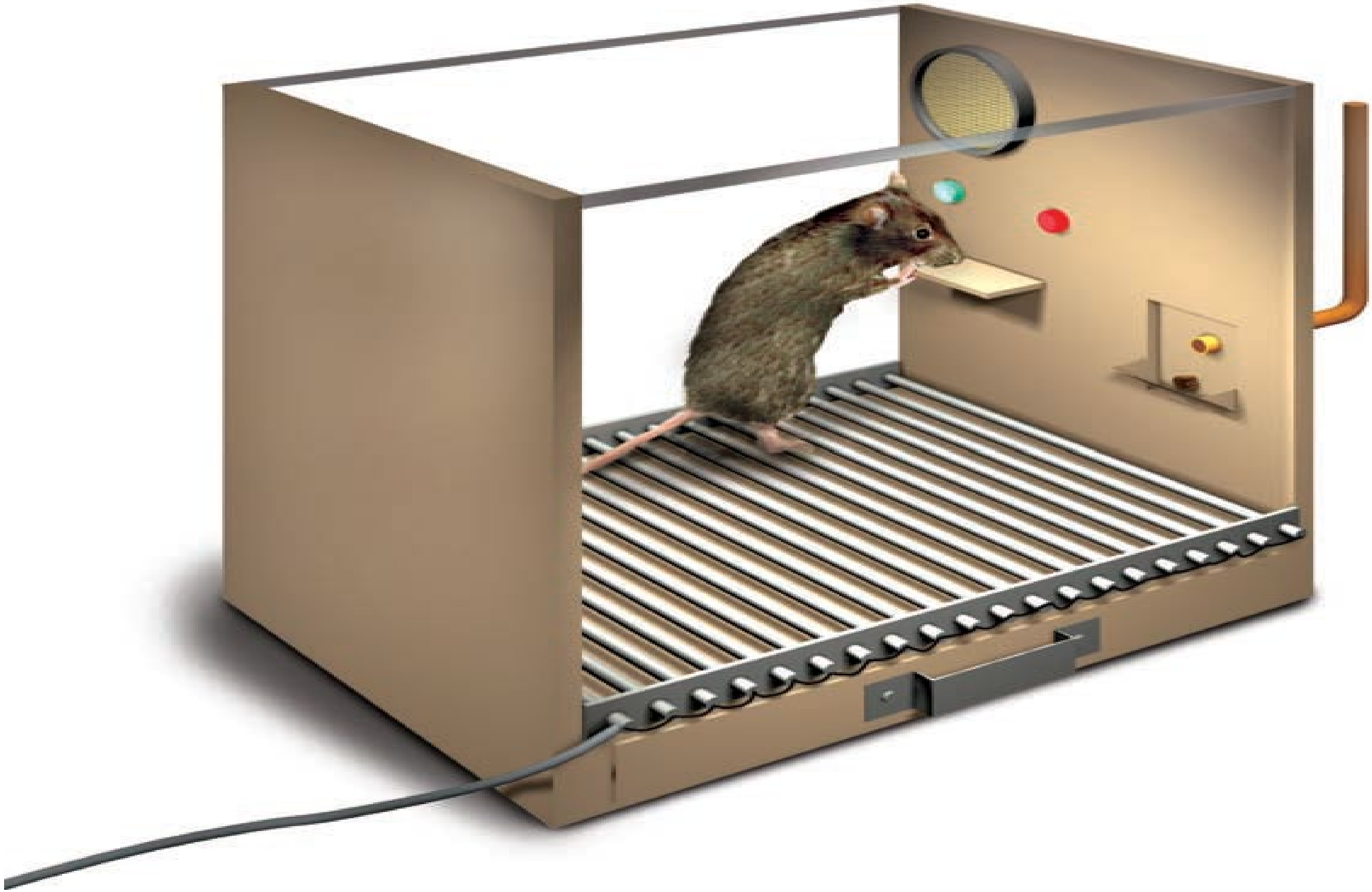
Examples

- **Example - 1**. Working industriously can bring about a raise in salary or bonus.

- **Example - 2**. Studying hard in college results in good academic grades.

- **Example - 3**. A patient moans and groans as he attempts to get up and walk for the first time after an operation. Praise and encouragement for his/her efforts will improve the chances that he/she will continue struggling toward independence.

Skinner's Experiment



- Skinner designed the Skinner box (Operant Chamber). An operant chamber is a simple box with a devise at one end that can be worked by the animal (rat, pigeon) in the box. According to the experiment, for the rats the devise is a lever. The lever is a switch that activates, when positive reinforcement is being used, a food delivery or water delivery mechanism. Thus, positive reinforcement is contingent(dependent) upon pressing a lever. Since this response is positively reinforced, it increases the frequency of pressing the lever.

Basic steps followed by Skinner in operant conditioning

1. A hungry animal (a cat, a rat, a pigeon) is placed in a laboratory setting (example - Skinner box)
2. The animal will wander in the box, exploring its environment in a random way.
3. The animal will press a lever by chance, which enables it to receive food from the food container.
4. The first time the response occurred, the animal will not learn the connection between lever pressing and the stimulus (food)
5. As the frequency of lever pressing increases, the animal learns that the receipt of food is dependent on lever pressing behavior.

In Skinner‘s Analysis,

- A response (operant) can lead to three types of consequences: such as
    A. A neutral consequence
    B. A reinforcement or
    C. Punishment

A. A neutral Consequence that does not alter the response.

Reinforcement

- A reinforcement that ***strengthens*** the response or makes it more likely to recur.
- A ***reinforcer*** is any event that ***increases*** the probability that the behavior that precedes it will be repeated.
- There are ***two*** basic types of reinforcers or reinforcing stimuli:
    - *Primary* reinforcers (naturally reinforcing because they satisfy biological needs & strengthen a behavior without prior learning)
    - *Secondary* reinforcers (reinforce behavior because of their prior association with primary reinforcing stimuli).

- Both primary and secondary reinforcers can be positive or negative.

Positive reinforcement

- is the process whereby ***presentation*** of a stimulus makes behavior more likely to occur again.

Negative reinforcement

- *is the process whereby **termination of an aversive stimulus** makes* behavior more likely to occur.
- The basic principle of negative reinforcement is that eliminating something aversive can itself be a reinforcer or a reward.

Schedules of Reinforcement

Continuous schedule of reinforcement

- When a response is first acquired, learning is usually most rapid if the response is reinforced each time it occurs.

Intermittent (partial) schedule of reinforcement

- which involves reinforcing only some responses, not all of them.
- response has become reliable, it will be more resistant to extinction if it is rewarded partially
- There are four types of intermittent schedules.
    1. Fixed-ratio schedules
    2. Variable-Ratio Schedule
    3. Fixed Interval Schedule
    4. Variable Interval Schedule

Ratio Schedules

Fixed-Ratio Schedules

- occurs after a *fixed number of responses*
- produce high rate of responding
- performance sometimes drops off just after reinforcement
- It is effective for motivating a great amount of work

Variable-Ratio Schedule

- occurs after *some average number of responses*
- produces extremely high steady rates of responding
- responses are more *resistant to extinction*

Interval Schedule

Fixed Interval Schedule

- reinforcement occurs only if a *fixed amount of time* has passed since the previous reinforcer.

Variable Interval Schedule

- reinforcement occurs only if *a variable amount of time* has passed since the previous reinforcer.

Examples

1. A man is paid after completing a certain amount of work.
2. Gambling games.
3. Payment of salary on 1st of every month.
4. Administration of quiz/test on every Monday of the class.
5. Fishing & Dialing a phone
6. Lowering the volume of the radio when you study prevents attention distraction.

Punishment

- is a stimulus that ***weakens*** the response or makes it less likely to recur.
- Punishers can be any ***aversive*** (unpleasant) stimuli that weaken responses or make them unlikely to recur.
- *Immediacy*, *consistency* and *intensity* matter are important for effectiveness of punishment.
- Like reinforcers, punishers can also be
    - Primary punishers (Pain and extreme heat or cold are inherently punishing
    - Secondary punishers (criticism, demerits, catcalls, scolding, fines, and bad grades)

Table 5.3 Four Ways to Modify Behavior

	Reinforcement	Punishment
POSITIVE (Adding)	Something valued or desirable; *Positive Reinforcement* Example: getting a gold star for good behavior in school	Something unpleasant; *Punishment by Application* Example: getting a spanking for disobeying
NEGATIVE (Removing/ Avoiding)	Something unpleasant; *Negative Reinforcement* Example: avoiding a ticket by stopping at a red light	Something valued or desirable; *Punishment by Removal* Example: losing a privilege such as going out with friends

Shaping

- is an operant conditioning procedure in which ***successive approximations*** of a desired response are reinforced.
- In shaping you start by reinforcing a tendency in the ***right direction***. Then you gradually require responses that are more and more similar to the final desired response.
- The responses that you reinforce on the way to the final one are called successive approximations.

Social Learning Theory (Observational Learning Theory)

- Developed by Albert Bandura
- Which is learning by watching the behavior of another person, or *model.*
- Learning rely on a ***social phenomenon***—it is often referred to as a *social cognitive approach to learning*
- Emphasis on interaction of behavior, environment, and person (cognitive) factors as determinants of learning
- Three ***forms*** of reinforcement that can encourage observational learning
  - *direct reinforcement*
  - *vicarious reinforcement*
  - *Self-reinforcement*

Cont...

Direct Reinforcement

- When the observer may reproduce the behaviors of the model and receive *direct reinforcement.*

Vicarious Reinforcement

- When the observer may reproduce the behaviors of the model but t*he reinforcement need not be direct - it may be vicarious reinforcement as* well.
- the observer may simply see others reinforced for a particular behavior and then increase his or her production of that behavior.

Self-Reinforcement

- *Or controlling your reinforcers.*
- *Important for* improve, value and enjoy their growing competence.

Four Conditions

Attention

- the person must first pay attention to the model.

Retention

- the observer must be able to remember the behavior that has been observed (rehearsal).

Motor reproduction

- the ability to replicate the behavior that the model has just demonstrated.

Motivation

- learners must want to demonstrate what they have learned.

Reading Assignment

- **Do video houses open at every corner of the city led teenagers to be delinquent in our society?**

Cognitive Learning Theory

- Cognitive processes are thus the *mental processes* involved in knowing about the world:- as such they are important in perception, attention, thinking, problem solving, and memory.
- Cognitive learning may take two ***forms***:
  - Latent learning
  - Insight learning (gestalt learning or perceptual learning)

Latent Learning

- Latent‘ means ***hidden***
- Is learning that occurs but is not evident in behavior until later, when ***conditions*** for its appearance are favorable.
- It occur without reinforcement of particular responses and seems to involve changes in the way information is processed.
- learning that is ***not immediately expressed***.
- A great deal of human learning also remains latent until ***circumstances*** allow or require it to be expressed.

Insight Learning

- It is a cognitive process whereby we ***reorganize our perception*** of a problem.
- In a typical insight situation where a problem is posed, a period follows during which no apparent progress is made, and then the solution comes suddenly.
- Human beings who solve a problem insightfully usually experience a good feeling called an ***'aha' experience.***
- Sometimes, for example, people even wake up from sleep with a solution to a problem that they had not been able to solve during the day.

CHAPTER FOUR

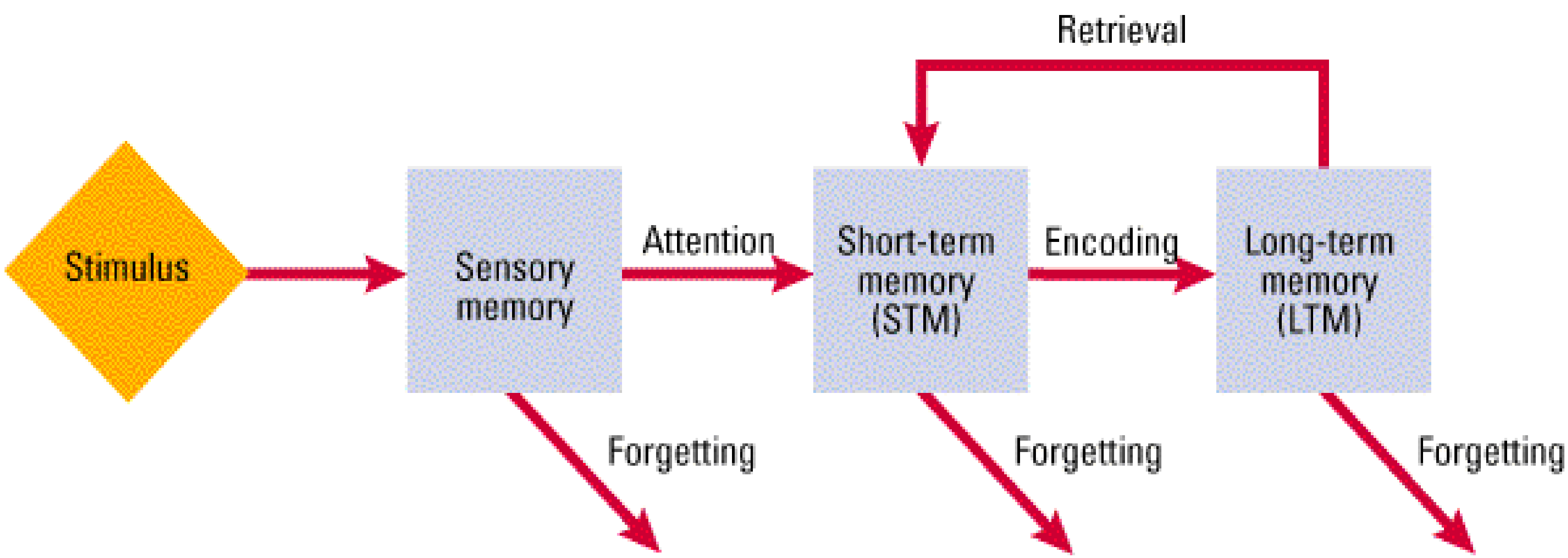
MEMORY AND FORGETTING

Brain storming Question

- What is the meaning of memory?
- What is the function of memory?
- What are the stage of memory model proposed by Atkinson and Shiffrin.
- Why do we call STM as a working memory?
- What is forgetting?
- How forgetting occur or what causes forgetting?

Meaning and Processes of Memory

- It is the *retention* of information/what is learned earlier over time.
- It is the way in which we record the past for later use in the present.
- Memory is a *blanket label* for a large number of processes that form the bridges between our past and



Processes of Memory

- It is the mental activities we perform to ***put*** information into memory, to ***keep*** it there, and to make ***use*** of it later.
- This involves three basic steps:
    a) Encoding
    b) Storage
    c) Retrieval

a) Encoding

- the term encoding refers to the ***form*** (i.e. the code) in which an item of information is to be ***placed*** in memory.
- the process by which information is ***initially recorded*** in a form usable to memory.
- In encoding we ***transform a sensory input into a form or a memory code*** that can be further processed.

b) Storage

- To be remembered the encoded experience must leave some record in the nervous system (***the memory trace***); it must be squirreled away and held in some more or less enduring form for later use.
- Storage is the ***persistence*** of information in memory.

c) Retrieval

- is the point at which one tries to ***remember*** to dredge up a particular memory trace from among all the others we have stored.
- In retrieval, material in memory storage is ***located***, brought into ***awareness*** and ***used***.

Memory is the process by which information is encoded (phase 1), stored (phase 2) and later retrieved (phase 3).

Stages/Structure of Memory

According to Atkinson and Shiffrin (1968), memory has three structures:

1. **Sensory Memory/Sensory Register**
2. **Short-Term Memory**
3. **Long Term Memory**

1) Sensory Memory/Sensory Register

- It is the ***entry*** way to memory (***first*** information storage area).
- It acts as a holding bin, retaining information until we can select items for attention.
- It gives us a ***brief time*** to decide whether information is extraneous or important.
- It can hold virtually ***all the information*** reaching our senses for a brief time.

Cont…

- For instance,
  - Visual images (Iconic memory) remain in the visual system for a maximum of ***one second***.
  - Auditory images (Echoic memory) remain in the auditory system for a slightly longer time, by most estimates up to ***two second*** or so.

- Information is accurate representation of the environmental information but ***unprocessed***.
- However, some of the information that has got attention and recognition **pass on short-term** memory for further processing.

Short-Term Memory

- It holds the ***contents*** of our attention.
- It consist memories of the ***by-products or end results of perceptual analysis.***
- Also called working memory, immediate memory, active memory, and primary memory.
- It has ***four*** characteristics:
  - It is active (workspace to process new information)there is consciously processing, examining, or manipulating information
  - Rapid accessibility: Information is readily available for use
  - Preserves the temporal sequence of information: maintain the information in sequential manner for a temporary period of time
  - Limited capacity: the magic number seven plus or minus

Chunking

- ***grouping or packing of information*** into higher order units that can be remembered as single units.
- It ***expands working memory*** by making large amounts of information more manageable.
- The real capacity of short-term memory, therefore, is not a few bits of information but a few chunks.

- STM memory holds information received from SM for up to ***about 30 seconds*** by most estimates.
- It is possible to prolong STM indefinitely by ***rehearsal***- the conscious repetition of information.
- Material in STM is easily displaced unless we do something to keep it there.

Long Term Memory

- It is a memory system used for the ***relatively permanent storage*** of ***meaningful information***.
- The capacity of LTM seems to have no practical limits.
- The vast amount of information stored in LTM enables us to learn, get around in the environment, and build a sense of identity and personal history.
- LTM stores information for ***indefinite periods***. It may last for days, months, years, or even a lifetime.

Sub Systems of LTM

- **Declarative/ explicit memory-** the conscious recollection of information such as specific facts or events that can be verbally communicated. Divided into two:
  - **Semantic memory-** factual knowledge like the meaning of words, concepts and our ability to do math. They are internal representations of the ***world***, independent of any particular context.
  - **Episodic memory**- memories for events and situations from personal experience. They are internal representations of ***personally*** experienced events.
- **Non-declarative/ implicit memory-** behavior is affected by prior experience without that experience being consciously recollected. One of the most important kinds of implicit memory is ***procedural memory***. It is the how to knowledge of procedures or skills: Knowing how to comb your hair, use a pencil, or swim.

Serial Position Effect

- The three-box model of memory is often invoked to explain interesting phenomenon called the serial position effect.
- If you are shown a list of items and are then asked immediately to recall them, your retention of any particular item will depend on its ***position in the list***.
- That is, recall will be best for items at the beginning of the list (***the primacy effect***) and at the end of the list (the ***recency effect***).
- When retention of all the items is plotted, the result will be a U-shaped curve.

Factors Affecting Memory

- **Ability to retain**: good memory traces left in the brain by past experiences.
- **Good health**: good health can retain the learnt material better
- **Age of the learner**: Youngsters can remember better than the aged.
- **Maturity**: Very young children cannot retain and remember complex material.
- **Will to remember**: Willingness to remember helps for better retention
- **Intelligence**: More intelligent person will have better memory
- **Interest**: will learn and retain better.
- **Over learning**: over learning will lead to better memory.
- **Speed of learning**: Quicker learning leads to better retention,
- **Meaningfulness of the material**: it remain in our memory for longer period
- **Sleep or rest**: after learning strengthens connections in the brain and helps for clear memory.

Forgetting

- The *apparent loss of information* already encoded and stored in the long-term memory.
- The first attempts to study forgetting were made by German psychologist *Hermann Ebbinghaus* (1885/1913).
- There is almost always a strong initial decline in memory, followed by a more gradual drop over time.
- Furthermore, *relearning* of previously mastered material is almost always faster than starting from a scratch

Theories of Forgetting

- Psychologists have proposed ***five mechanisms*** to account for forgetting:

1) The Decay Theory
2) Replacement of old memories by new ones
3) Interference
4) Motivated forgetting
5) Cue dependent forgetting

The Decay Theory

- memory traces or engram fade with time if they are ***not accessed now and then***.
- In decay, the trace simply fades away with nothing left behind, because of the ***passage of time***.

Interference Theory

- It occurs because *similar items of information interfere with one another* in either storage or retrieval.
- There are two kinds of interference :
  - In **Proactive Interference**, information learned earlier interferes with recall of newer material.
  - If new information interferes with the ability to remember old information the interference is called **Retroactive Interference.**

New Memory for Old/ Displacement Theory

- This theory holds that ***new information entering memory can wipe out old information***, just as recording on an audio or videotape will obliterate/wipe out the original material.

Motivated Forgetting

- Sigmund Freud maintained that people forget because they block from consciousness those memories that are ***too threatening or painful to live with***, and he called this self-protective process Repression.

Cue Dependent Forgetting

- When we ***lack retrieval cues***, we may feel as if we have lost the call number for an entry in the mind‘s library.
- In long-term memory, this type of memory failure may be the most common type of all.

Improving Memory

- **Pay Attention**: It seems obvious, but often we fail to remember because we never encoded the information in the first place.
- **Encode information in more than one way**: The more elaborate the encoding of information, the more memorable it will be
- **Add meaning**: The more meaningful the material, the more likely it is to link up with information already in long-term memory.
- **Take your time**: If possible, minimize interference by using study breaks for rest or recreation. Sleep is the ultimate way to reduce interference.
- **Over learn**: Studying information even after you think you already know it- is one of the best ways to ensure that you‘ll remember it.
- **Monitor your learning**: By testing yourself frequently, rehearsing thoroughly, and reviewing periodically, you will have a better idea of how you are doing.

Critical Questions

- What were the main assumptions of the Atkinson-Shiffrin model (1971)? How did they describe the process of memory?
- According to Baddeley (2001), what the four main components of working memory and Explain it?
- The text states that forgetting is due to both decay and interference. Do you feel like one might play a bigger role than the other? Why?
- According to theories of independent memory systems, what are the various memory systems that are distinguished primarily by the types of information they handle and explain it?
- What do synaptic transmission and hormonal fluctuations have to do with memory?

Chapter Five
Motivation And Emotion

Brainstorming

- Why do some people run after money and some refuse even the most attractive job offers?
- Why do some people leave their country for earning money and some are contented and happy with whatever is available to them at home?
- Why people become doctors, accountants, engineers, social workers, pilots, army men etc?

Definition and Types of Motivation

- It is a factor by which activities are started, directed and continued so that physical or psychological needs or wants are met.
- The word itself comes from the Latin word 'Mover', which means -to move.
- Motivation is what -moves people to do the things they do.

Types of motivation.

- **Intrinsic motivation** is a type of motivation in which a person acts because the act itself is rewarding or satisfying in some internal manner.
- **Extrinsic motivation** is a type of motivation in which individuals act because the action leads to an outcome that is external to a person.

Approaches to Motivation

- The sources of motivation are different according to the different theories of motivation.
- There are many causes of behaviour. People perform behaviour for a number of reasons.
- Psychologists have been studying the causes of behaviours and have developed various theories that explain the why of these behaviours
- Some of these theories are
  - Instinct approaches
  - Drive-reduction approaches
  - Arousal approaches
  - Incentive approaches
  - Cognitive approaches
  - Humanistic approaches

Instinct Approaches to Motivation

- This theory states that motivation is the result of an inborn, biologically determined pattern of behavior.
- According to this approach, people and animals are born with programmed sets of behavior essential to their survival.
- Motivation is evolutionarily programmed through inborn instinctual behavior patterns.
- According to this instinct theory, in humans, the instinct to reproduce is responsible for sexual behavior, and the instinct for territorial protection may be related to aggressive behavior.
- One important thing by forcing psychologists to realize that some human behavior is controlled by hereditary factors.

Drive-Reduction Approaches

- This approach involved the concepts of needs and drives.
- A need is a requirement of some material (such as food or water) that is essential for the survival of the organism.
- When an organism has a need, it leads to a psychological tension as well as physical arousal to fulfill the need and reduce the tension. This tension is called *drive*.
- It proposes just this connection between internal psychological states and outward behavior.
- In this theory, there are two kinds of drives:
  - **Primary drives** (biological needs)are those that involve survival needs of the body such as hunger and thirst,
  - **Secondary drives** (acquired drives) are those that are learned through experience or conditioning, such as the need for money, social approval.

Cont…

- Behavior is motivated by desires to reduce internal tension caused by unmet biological needs, such as hunger or thirst.
- Internal drives “push” us to behave in certain ways.
- Robert Woodworth & Clark Hull: Drives are triggered by internal mechanisms of *homeostasis*→ Internal state of balance.
- Limitations: All behaviors are not always motivated purely by physiological needs.

Arousal Approaches

- Explain behavior in which the goal is to maintain or increase excitement.
- People take certain actions to either decrease or increase levels of arousal.
- People are motivated to maintain an *optimal level of arousal*
- Each person tries to maintain a certain level of stimulation and activity.
- As with the drive-reduction model, this approach suggests that if our stimulation and activity levels become too high, we try to reduce them.
- But, in contrast to the drive-reduction perspective, the arousal approach also suggests that if levels of stimulation and activity are too low, we will try to increase them by seeking stimulation.

Incentive Approaches

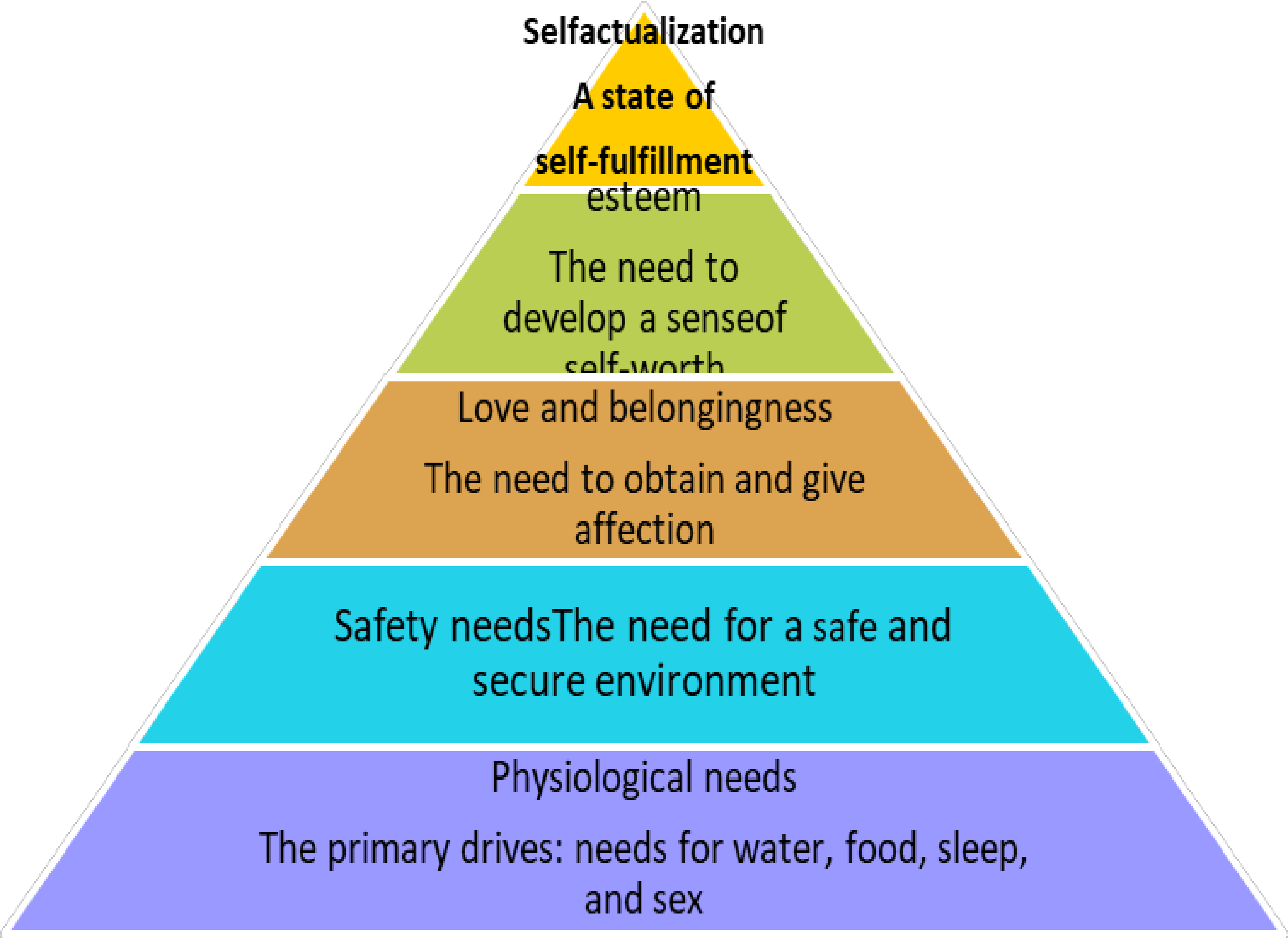
- suggest that motivation stems from the desire to attain external rewards.
- The desirable properties of external stimuli: account for a person's motivation.
- Embedded in behavioral learning concepts as association and reinforcement.
- Many psychologists believe that the internal drives proposed by drive-reduction theory work in a cycle with the external incentives of incentive theory to 'push' and 'pull' behavior, respectively.
- Hence, at the same time that we seek to satisfy our underlying hunger needs (the push of drive-reduction theory), we are drawn to food that appears very appetizing (the pull of incentive theory).
- Rather than contradicting each other, then, drives and incentives may work together in motivating behavior.

Cognitive Approaches

- Suggest that motivation is a result of people's thoughts, beliefs, expectations, and goals.
- Cognitive theories of motivation draw a key difference between intrinsic and extrinsic motivation.
- Intrinsic motivation causes us to participate in an activity for our enjoyment rather than for any actual or concrete reward that it will bring us.
- In contrast, extrinsic motivation causes us to do something for money, a grade, or some other actual, concrete reward.

Humanistic approaches to motivation

- Maslow suggested that human behavior is influenced by a hierarchy, or ranking, of five classes of needs, or motives.
- Maslow's Hierarchy of Needs.
- People have strong cognitive reasons to perform various actions.
- Motivation to realize their highest personal potential.
- He said that needs at the lowest level of the hierarchy must be at least partially satisfied before people can be motivated by the ones at higher levels.
- Maslow‘s five Hierarchies of needs for motives from the bottom to the top are as follows:


Selfactualization
A state of
self-fulfillment
esteem
The need to
develop a senseof
self-worth
Love and belongingness
The need to obtain and give
affection
Safety needsThe need for a safe and
secure environment
Physiological needs
The primary drives: needs for water, food, sleep,
and sex

Conflict of Motives and Frustration

- Based on the sources of motivation and the importance of the decision, people usually face difficulty choosing among the motives.
- These are just a few of the motives that may shape a trivial decision.
- When the decision is more important, the number and strength of motivational pushes and pulls are often greater, creating far more internal conflict and indecision.
- refers to negative emotional state (depression, anger, anxiety, etc) that develop when a person is unable to make a choice between two or more alternatives.
- There are four basic types of motivational conflicts.
    - Approach-approach conflicts
    - Avoidance-avoidance conflicts
    - Approach-avoidance conflicts
    - Multiple approach-avoidance conflicts

Cont…

Approach-approach conflicts

- exist when we must choose only one of the two desirable activities.

Avoidance-avoidance conflicts

- arise when we must select one of two undesirable alternatives.

Approach-avoidance conflicts

- happen when a particular event or activity has both attractive and unattractive features

Multiple approach-avoidance conflicts

- exist when two or more alternatives each have both positive and negative features.

- Suppose you must choose between two jobs. One offers a high salary with a well-known company but requires long working hours and relocation to a miserable climate. The other boasts advancement opportunities, fringe benefits, and a better climate, but it doesn‘t pay as much and involves an unpredictable work schedule.
- An individual may be torn between the idea of going to a political rally or a movie which he likes to do equally.
- Such conflicts are capsuled in the saying “ caught between the devil and the deep blue sea “
- The closer you are to something appealing, the stronger your desire to approach it, the closer you are to something unpleasant, the stronger your desire to flee.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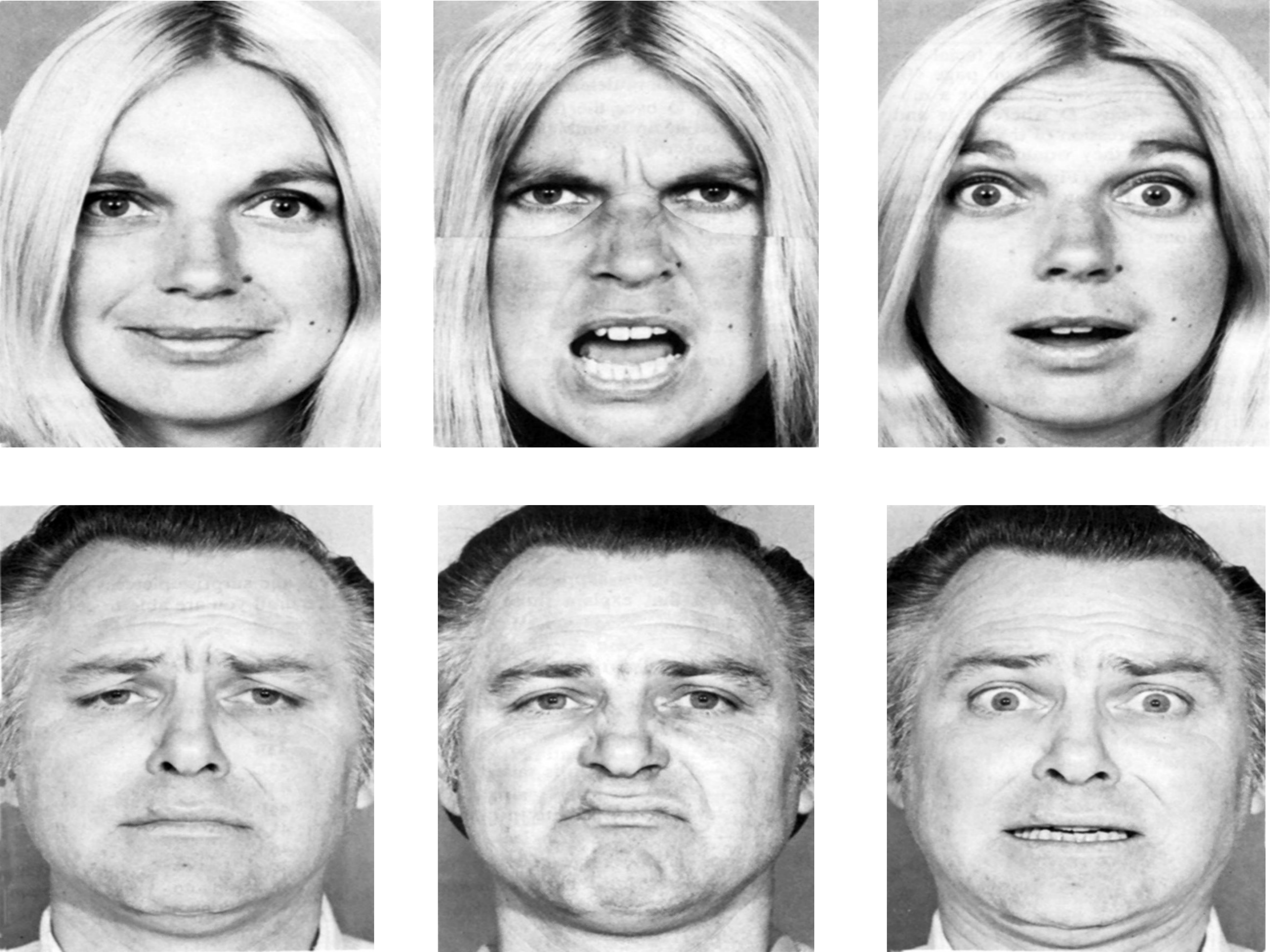
2
3



Emotions

Definition of Emotion

- Latin "*Emovere*" → "To excite, stir up or agitate."
- It refers to a strong feeling about something.
- the 'feeling' aspect of consciousness, characterized by certain physical arousal, certain behavior that reveals the feeling to the outside world, and an inner awareness of feelings.
- Emotions are feelings such as happiness, despair, and sorrow that generally have both physiological and cognitive elements influencing behavior.
- While motives are internally caused, emotions are responses to an external stimulus.

Three Components of Emotion

The physiology of emotion

- when a person experiences an emotion, there is physical arousal created by the sympathetic nervous system.
- Bodily arousal: Increased heart rate, blood pressure, outward blood flow, activity of the stomach and gastro intestinal system, hormonal increase, respiration, etc.
- ANS: Consists of two parts:

  - *Sympathetic Nervous System*: Activated in response to external threats and arouses the body for action.
  - *Parasympathetic Nervous System*: Supports activities that maintain the body to restore energy.

- It calms down the body to maintain energy by slowing heart rate, lowering blood pressure, and so on.

Cont…

The behavior of emotion

- tells us how people behave in the grip of an emotion.
- Characteristic overt expressions of emotions.
- There are facial expressions, tone of voice, touching, posture, Body gestures, body movements, and actions that indicate to others how a person feels.

Subjective experience or labeling emotion

- it involves interpreting the subjective feeling by giving it a label: anger, fear, disgust, happiness, sadness, shame, interest, surprise and so on.
- Subjective conscious experience and interpretation of emotions.
- The thoughts, beliefs & expectations determining the type and intensity of the emotional response.

Theories of Emotions

- Major theories of emotion are grouped into three:
  - **Physiological**: Bodily responses are responsible for emotions.
  - **Neurological**: Brain activity leads to emotional responses.
  - **Cognitive**: Mental processes play an essential role in formation of emotions.

Theories of Emotion

James- Lang Theory of Emotion

- This theory of emotion is based on the work of William James & Carl Lang (1885).
- In this theory, a *stimulus* of some sort (for example, the large snarling dog) produces a physiological reaction. This reaction, which is the *arousal* of the ‗fight-or-flight‖ sympathetic nervous system (wanting to run), produces bodily sensations such as increased heart rate, dry mouth, and rapid breathing.
- James and Lang believed that physical arousal led to the labeling of the *emotion* (fear).

Cannon-Bard theory of emotion

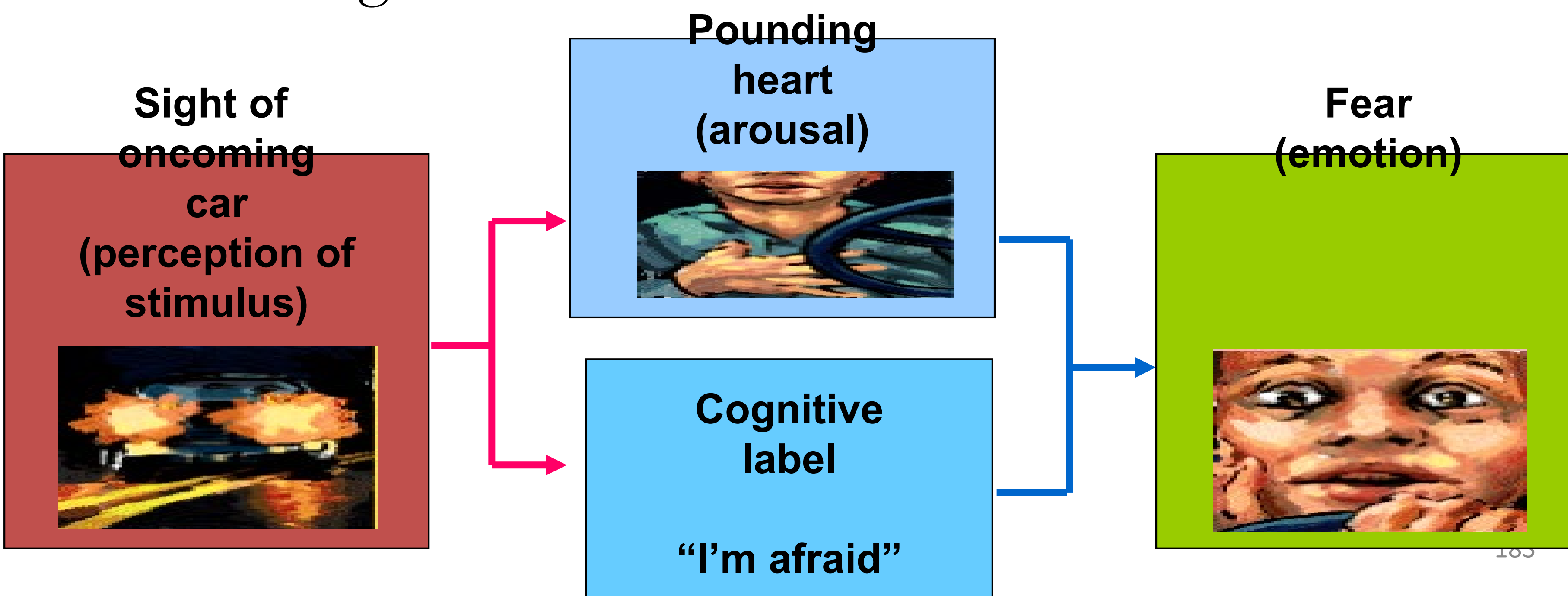
- Developed by Physiologists Walter Cannon and (1927) and Philip Bard (1934) .
- theorized that the emotion and the physiological arousal occur more or less at the same time.
- Cannon, an expert in sympathetic arousal mechanisms, did not feel that the physical changes aroused by different emotions were distinct enough to allow them to be perceived as different emotions.

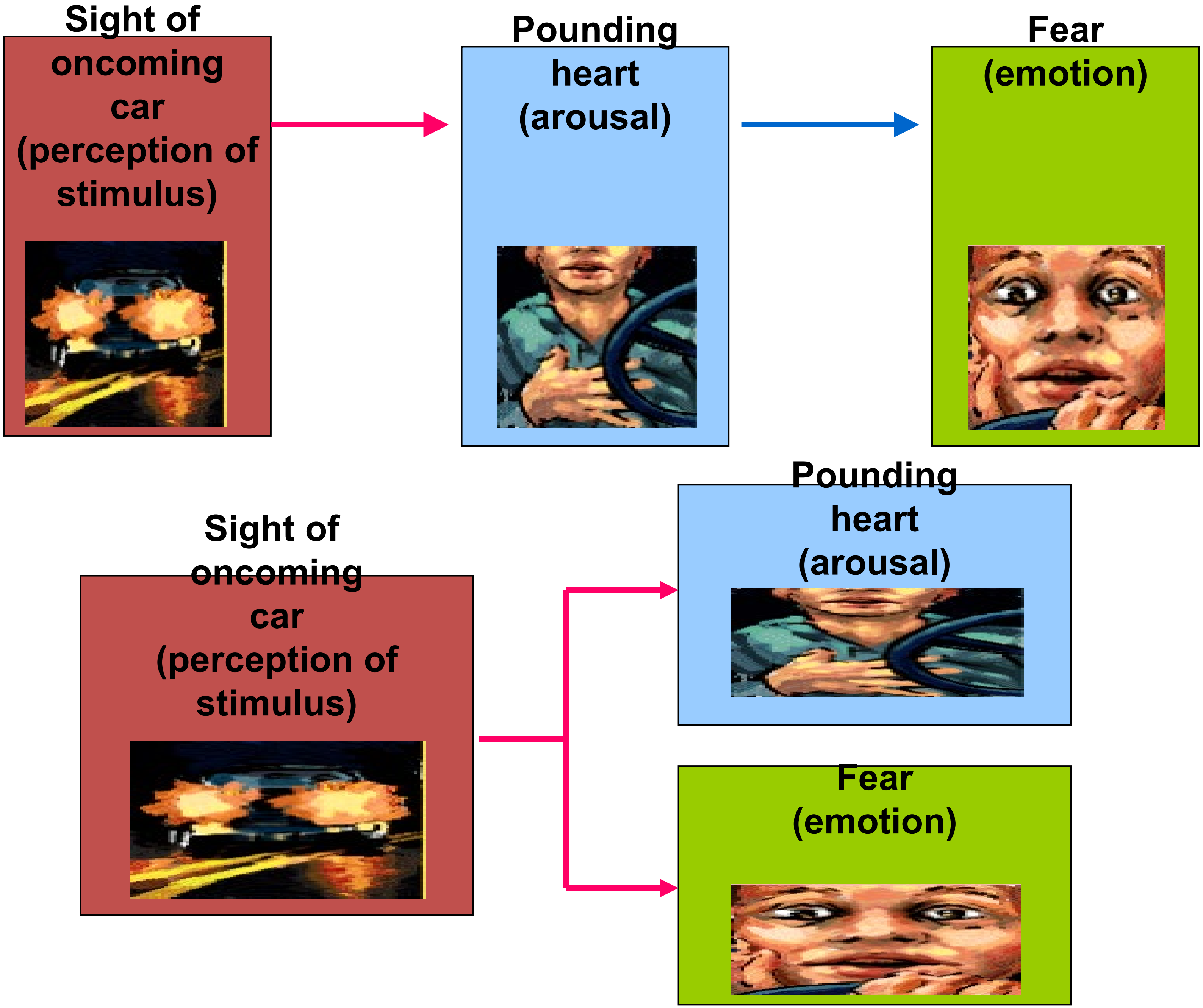
- Bard expanded on this idea by stating that the sensory information that comes into the brain is sent simultaneously (by the thalamus) to both the cortex and the organs of the sympathetic nervous system. The fear and the bodily reactions are, therefore, experienced at the same time-not one after the other.

Cont…

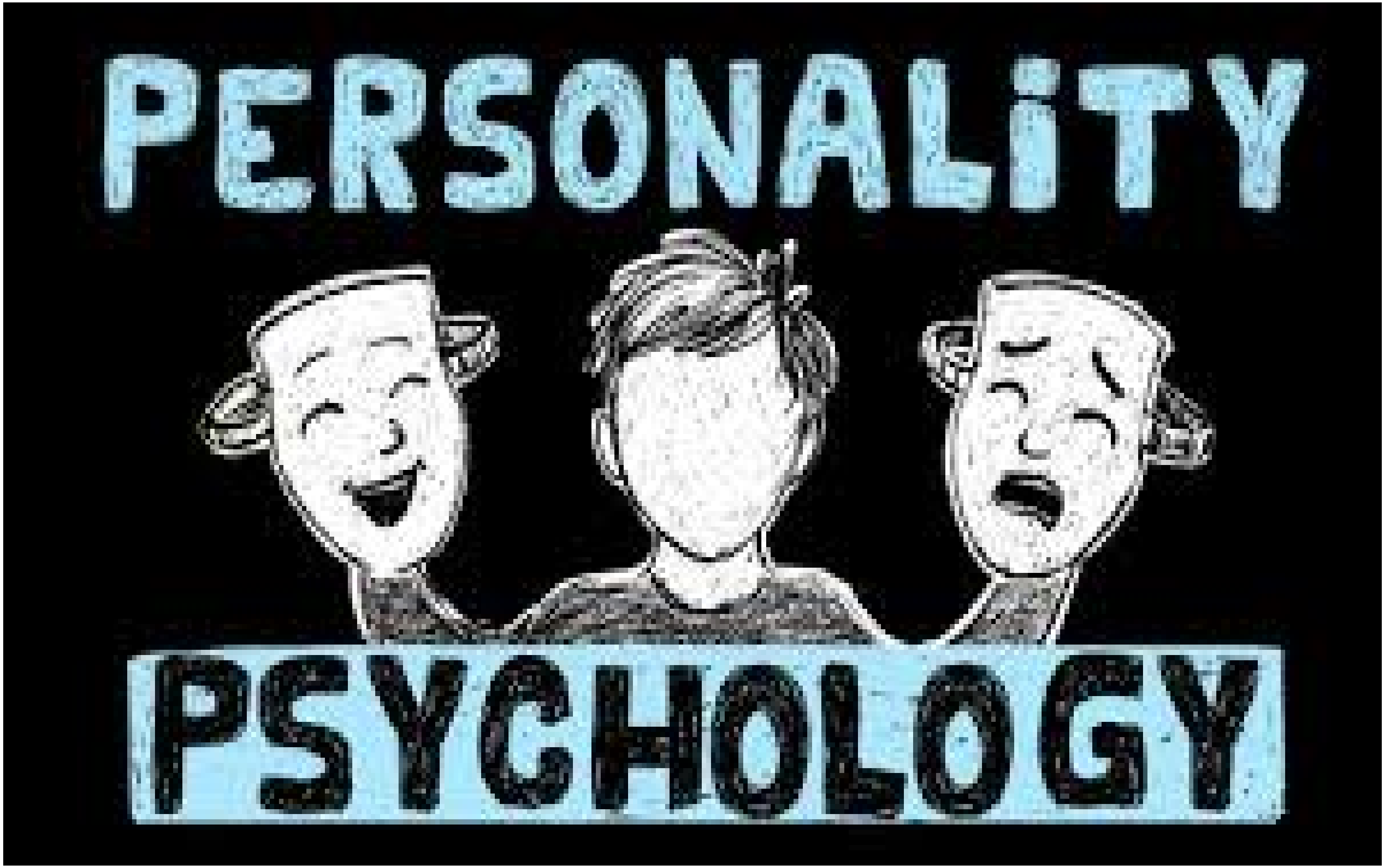
Schechter-Singer and Cognitive Arousal Theory

- They proposed that two things have to happen before emotion occurs: the physical arousal and labeling of the arousal base on cues from the surrounding environment.
- These two things happen at the same time, resulting in the labeling of the emotion.




Sight of oncoming car (perception of stimulus)
Pounding heart (arousal)
Fear (emotion)
Sight of oncoming car (perception of stimulus)
Pounding heart (arousal)
Fear (emotion)

CHAPTER SIX
PERSONALITY




extravagant
eloquent
cunning
down-to-earth
competitve
trustful
candid
bossy
considerate
affectionate
high-strung
persistent
mischeivous
self-satisfied
immodest
sensitive
patient
unpretentious
reliable
extraverted
simple
vigorous
jovial
flamboyant
boastful
uncritical
friendly
adventurous
mannerly
exhibitionist
loyal
sociable
romantic
excitable
dominant
assertive
kind
warm
rough
helpful
energetic
moral
active
crude
servile
unscrupulous
gullible
firm
explosive
sincere
assured
witty
communicative
manipulative
flirtatious
happy
brave
social
daring
cordial
wordy
bullheaded
bold
forward
optimistic
intense
proud
uninhibited
tolerant
fair
peaceful
merry
boisterous
dramatic
reckless
volatile
cheerful
verbal
unrestrained
respectful
expressive
abrupt
mature
generous
magnetic
talkative
sly
spirited
forceful
verbose
worldly
easy-going
pompous
soft
pleasant
flexible
tactful
alert
unruly
polite
genial
aggressive
enthusiastic
outspoken
charitable
deep
idealistic
gregarious
independent
strong
vibrant
zestful
confident
sentimental
emotional
combative
sympathetic
effervescent
relaxed
humorous
courteous
inquisitive
compassionate
spontaneous
understanding
courageous
domineering
opportunistic
reasonable
unselfconscious
cooperative
opinionated
demonstrative
ambitious
dependent
diplomatic
conceitless
conscientious

Meaning of Personality

- Derived from Latin word **"PERSONA"**, theatrical ***masks*** worn by Greek actors.
- The **unique pattern** of enduring thoughts, feelings, and actions that characterize a person.
- Terms:
    - » Character (moral or ethical behavior)
    - » Temperament (enduring characteristics)

Determinants of Personality

- Heredity/ Physiological Determinants
  - physical differences, IQ, potential, temperament
- Environment
  - culture, norms of family, friends & social groups, other influences
- Situation
  - in class vs. at a party
  - on-the-field/court vs. off-the-field/court

Theories of Personality

Three major theories of personality

- The psychoanalytic theory of personality
- The trait theory of personality
- Humanistic theory of personality

- Each of these [perspectives on personality] attempts to describe different patterns in [personality], including how these patterns form and how people differ on an individual level.

Psychoanalytic Theory

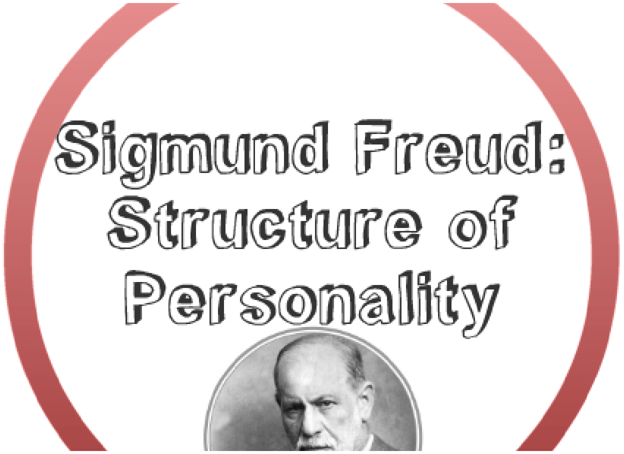
- By the Austrian physician named ***Sigmund Freud***.

Assumption

- personality is formed within ourselves, arising from basic inborn needs, drives, and characteristics.
- He argued that people are in constant conflict between their biological urges (drives) and the need to tame them.
- focuses on the **unconscious intra-psychic dynamics**
- focuses in the importance and primacy of the **first five years of life (early childhood influences).**
- a result of conflict between different **personality** systems or structures
- an interaction between nature (innate instincts) and nurture (parental influences).

Personality Structures

» Id
» Ego
» Superego



Assumption

- Which serves a different *function* and develops at different *times*
- *Interaction*: one another determines the personality of an individual.
- Any actions we take or problems we have results from the *interactions or degree of balance* among these systems.

Id: If It Feels Good, Do It

- *First* and most *primitive* part (infant/at birth)
- The Id is a Latin word that means "it"
- *Unconscious* amoral part of the personality
- *Libido*: instinctual energy
- Guided by *pleasure principle* (immediate satisfaction of needs).
- Containing all of the *basic biological drives; hunger, thirst, sex, aggression*
- It is *oblivious* to rules and regulations

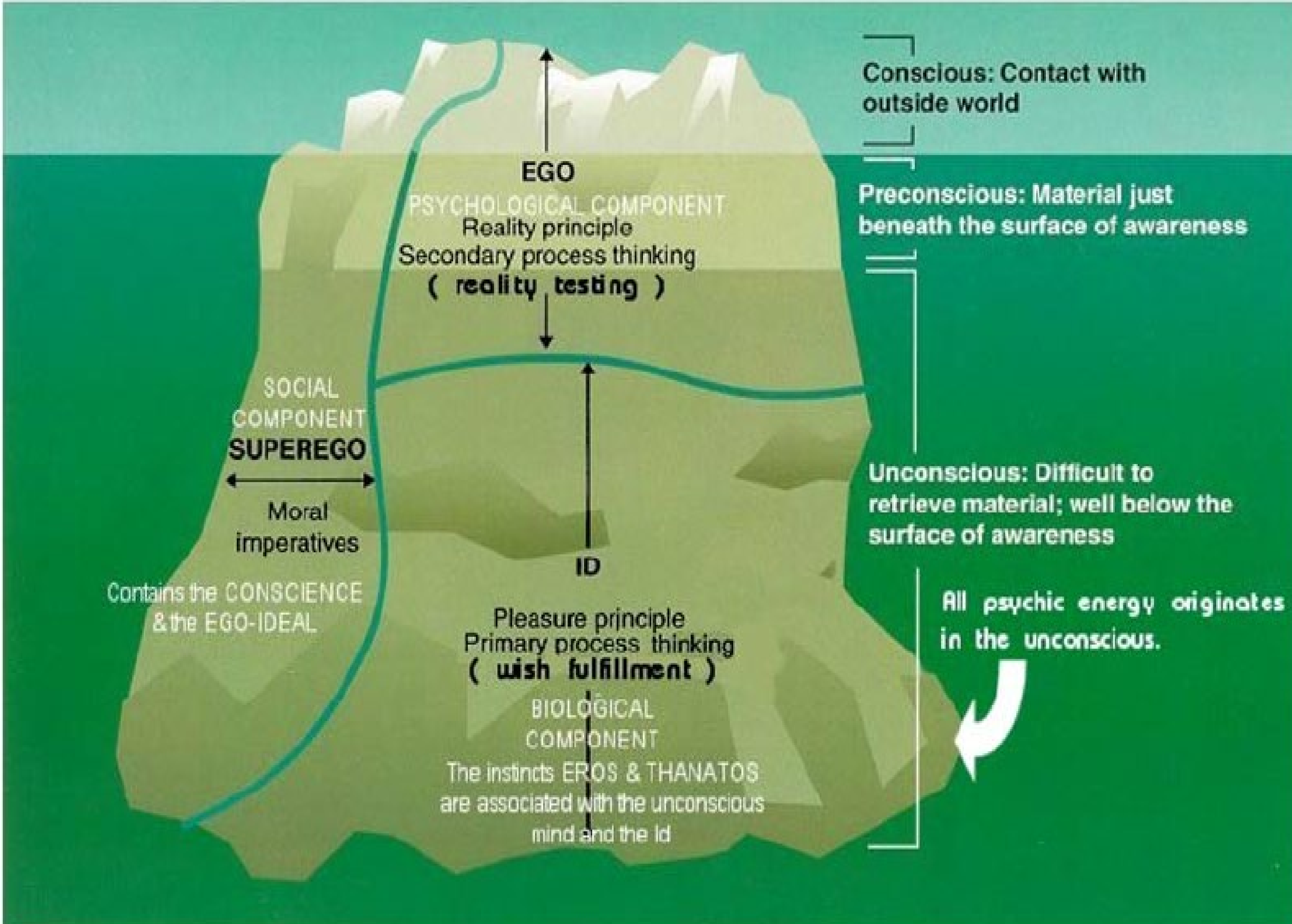
Ego: The Executive Director

- *Second* part of personality
- Latin word for "I",
- Mostly *conscious* and is far more *rational*, logical and cunning than the id
- Guided by *reality principles* (balance between libido & reality)
- Serves as a *referee or mediator*

Superego: The Moral Watchdog

- The *third* and final part of the personality
- Latin word, meaning *"over the self"*
- Guided by *moral principles*
- Includes all the moral codes (rules, customs, and expectations) of society
- It aspires for and expects the individual to be *perfect*.
    - *Ego-ideal* is a kind of measuring device (correct and acceptable behavior)
    - *Conscience* -that makes people pride/satisfaction/mental peace when they do the right thing and guilt/shame/self-blame, or moral anxiety when they do the wrong thing.

Freud's model of personality structure



Defense Mechanisms

- For Freud, our personality is the outcome of the continual *battle* for dominance among *the id, the ego, and the superego*.
- This constant conflict between them is managed by psychological defense mechanisms.
  - *unconscious tactics* that either prevent threatening material from surfacing or disguise it when it does.
  - refer to methods used by the ego to *prevent anxiety or threatening thoughts .*
  - Are useful to reduce anxiety and make us *feel normal* again.
  - They only become harmful if or when they are used *excessively*.
  - In order to justify one's action which is wrong in the eyes of the superego, the ego has to deny, distort or twist the reality.

Repression

– ***banishing*** threatening thoughts, feelings, and memories into the ***unconscious*** mind.

Denial

– ***refusal*** to recognize or acknowledge a threatening situation.

Regression

– involves reverting to ***immature*** behaviors that have relieved anxiety in the past.

Rationalization

– giving ***socially acceptable*** reasons for one's inappropriate behavior.

Displacement

- involves expressing feelings toward a person who is ***less threatening*** than the person who is the true target of those feelings.

Projection

- involves ***attributing*** one's undesirable feelings to other people.

Reaction formation

- involves a tendency to act in a manner ***opposite*** to one's true feelings.

Sublimation

- involves expressing sexual or aggressive behavior through indirect, socially ***acceptable*** outlets.

- After cheating someone in a business transaction, you might reduce your guilt by rationalizing that “everyone does it.”
- If you forget a dental appointment or the name of someone you don’t like.
- If lusting for a co-worker makes you feel guilty, you might attribute any latent sexual tension between the two of you to the *other person’s* desire to seduce you.
- **If** your boss gives you a hard time at work and you come home and slam the door, kick the dog, and scream at your spouse, you’re displacing your anger onto irrelevant targets.

- A fired executive having difficulty finding a new job might start making ridiculous statements about his incomparable talents and achievements.
- A student watches TV instead of studying, saying that "additional study behavior wouldn't do any good anyway."
- After parental scolding, a young girl takes her anger out on her little brother.
- A woman who dislikes her boss thinks she likes her boss but feels that the boss doesn't like her.
- A traumatized soldier has no recollection of the details of a close brush with death.

The Trait Theory of Personality

- It is a combination of ***stable internal characteristics*** that people display ***consistently over time and across situations***.

Assumptions

- Personality traits are **relatively stable**, across situations, and therefore ***predictable***, over time.
- People differ in ***how much*** of a particular personality trait they possess; no two people are exactly alike on all traits.
- The result is an endless variety of ***unique*** personalities.
- Personality is biologically based

The Big five Theory

- Robert McCrae and Paul Costa
- is also known as the five factor model
- is known as OCEAN for short

Cont…

Openness

- a person's willingness to try ***new things*** and be ***open to new experiences.***
- curiosity , flexibility and imaginative tendency

Conscientiousness

- a person's ***organization*** and motivation
- are ***careful*** about being in places on time and careful with belongings as well.
- dependability and responsibility of the individual

Extraversion

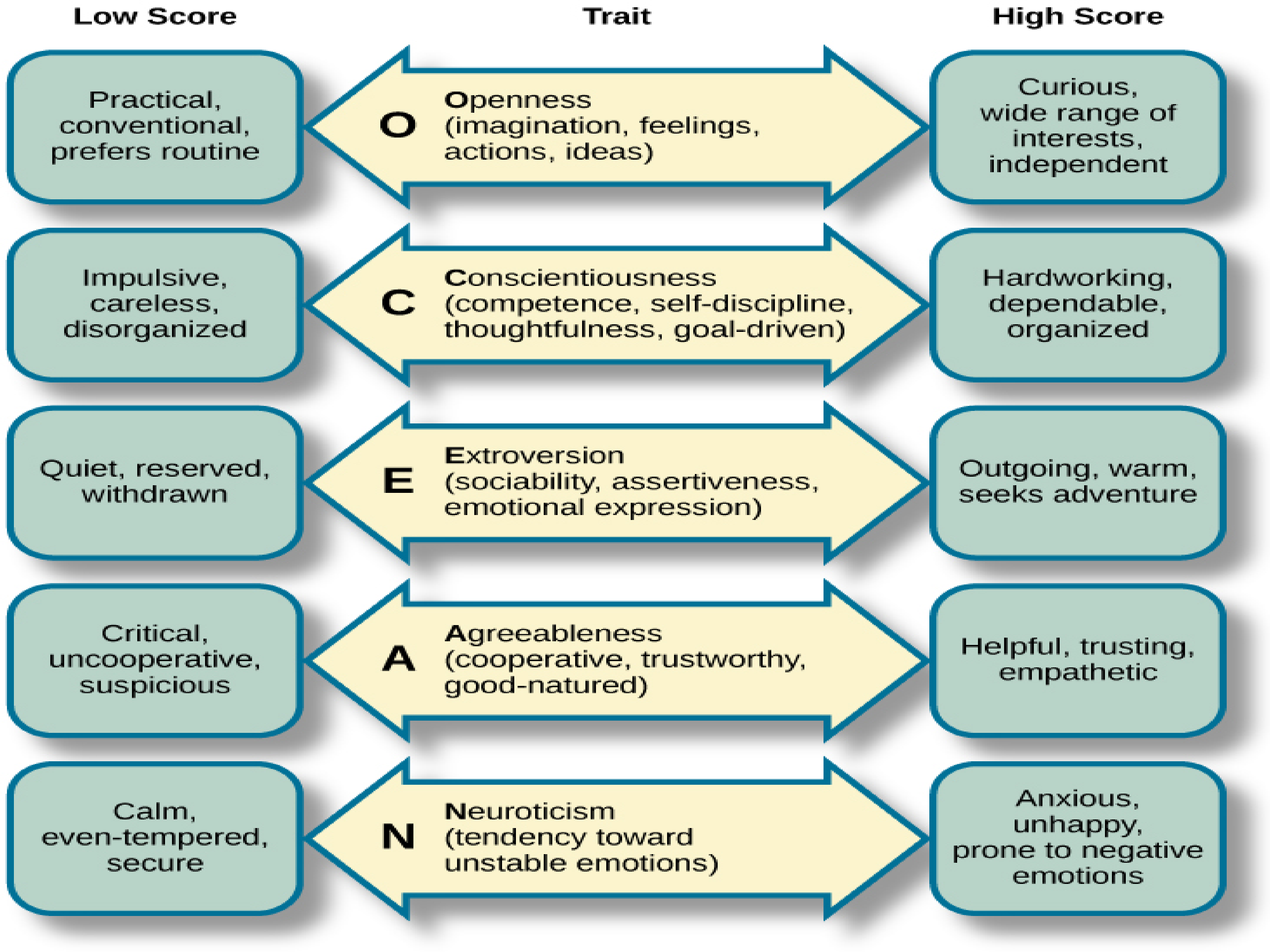
- all people could be divided into two personality types: extraverts and introverts (***Carl Jung***).
- Extraverts are outgoing , sociable, fun-loving, whereas introverts are more solitary and dislike being the center of attention

Agreeableness

- refers to the basic emotional style of a person, who may be easygoing, friendly, helpful, cooperative and pleasant (at the high end of the scale) or grumpy, crabby, hostile, self centered and hard to get along with (at the low end).

Neuroticism

- refers to emotional instability or stability.
- People who are excessively worried, overanxious and moody would score high on this dimension, whereas those who are more even-tempered and calm could score low.


Low Score
Trait
High Score
Practical, conventional, prefers routine
O
Openness (imagination, feelings, actions, ideas)
Curious, wide range of interests, independent
Impulsive, careless, disorganized
C
Conscientiousness (competence, self-discipline, thoughtfulness, goal-driven)
Hardworking, dependable, organized
Quiet, reserved, withdrawn
E
Extroversion (sociability, assertiveness, emotional expression)
Outgoing, warm, seeks adventure
Critical, uncooperative, suspicious
A
Agreeableness (cooperative, trustworthy, good-natured)
Helpful, trusting, empathetic
Calm, even-tempered, secure
N
Neuroticism (tendency toward unstable emotions)
Anxious, unhappy, prone to negative emotions

Humanistic theory of personality

(Carl Rogers, Abraham Maslow)

- ✓ Emphasize people's ***inherent goodness and their tendency to move toward higher levels of functioning*** instead of seeing people as controlled by the unconscious.
- ✓ Assume people have *conscious*, ***self-motivated ability*** to change and improve, along with people's unique creative impulses
- ✓ is regarded as the ***third force*** in psychology

Carl Rogers and Self-Concept

Assumption

- human beings are always ***striving*** to fulfill their innate capacities and capabilities and to become everything that their ***genetic potential will allow*** them to become.
- This striving for fulfillment is called ***self-actualizing tendency.***
- An important tool in human self-actualization is the development of an image of oneself or the ***self-concept.***
- The self-concept is based on what people are told by ***others*** and how the sense of self is reflected in the *words and actions* of important people in one's life, such as parents, siblings, coworkers, friends, and teachers.

Cont…

- ✓ The self concept is an important element in this theory
  - ❖ The real self
    - One‘s actual perception of characteristics, traits, and abilities that form the basis of the striving for self-actualization
  - ❖ The ideal self
    - ✓ The perception of what one should be or would like to be
- ✓ Rogers believed that when the real self and the ideal self are very close or similar to each other, people feel competent and capable. Otherwise anxiety and neurotic behavior can be the result.

Conditional and Unconditional Positive Regard

- Conditional positive regard?
- Unconditional positive regard ?

- ***How do you explain the importance of positive regard for personality development?***

  – positive regard as warmth, affection, love, and respect that comes from the significant others in people's experience.
  – Positive is vital to people's ability to cope with stress and to strive to achieve self-actualization.

Cont…

- Rogers believed that unconditioned positive regard, **or love, affection and respect with no strings attached, is** necessary for people to be able to ***explore fully all that they can achieve and become.***
- Unfortunately, some parents, spouses, and friends give conditional positive regard**, which is** love, affection, respect and warmth that depend, or seem to depend, on doing what those people want.
- For Rogers, a person who is in the process of self-actualizing, activity exploring potentials and abilities and experiencing a match between real and ideal selves is **a fully functioning person**.

Carl Rogers

Carl Rogers: American psychologist; believed that personality formed as a result of our strivings to reach our full human potential.

Fully Functioning Person: Lives in harmony with his/her deepest feelings and impulses

Self-Image: Total subjective perception of your body and personality

Conditions of Worth: behaviors and attitudes for which other people, starting with our parents, will give us positive regard.

Unconditional Positive Regard: Unshakable love and approval

Positive Self-Regard: Thinking of oneself as a good, lovable, worthwhile person